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The Literature Review

A Step-by-Step Guide for Students

Diana Ridley

Second Edition



The Literature Review

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1

Introduction

'The first thing you must do is conduct a comprehensive literature review.' This is one of the early tasks that many research students are asked to undertake by their supervisors. It can sometimes be frustrating, when you are keen to make a start on researching a particular problem that you have identified, to be advised to go and search for relevant work done by others, to identify pertinent theories, or just generally to read around the subject. However, it *is* important to explore the field in which you are going to do your research and gain a thorough awareness and understanding of current work and perspectives in the area so that you can position your own research clearly on the academic map of knowledge creation. An essential aspect of academic research is that it has connections with the work of others.

'But where do I start?' and 'What exactly is a literature review, anyway?' are common questions. The literature review is an activity which has many facets and it is not unusual to feel somewhat baffled when trying to work out exactly what is expected. To be tackled successfully, it requires thoughtful organisation and planning from the beginning. Before embarking on the challenge it is therefore helpful to give careful consideration to the following questions in the context of your own discipline:

- What is a literature review and what purposes does it serve in relation to your research?
- Why is it such a vital component of research?
- What are the various ways it can be realised in a dissertation or thesis?
- What does the process of completing a literature review involve?

I begin to address these questions here and elaborate further on the ideas throughout later chapters. The book as a whole contains:



- extensive practical tips on how to prepare, organise and write your review successfully;
- extracts from a variety of different sample dissertations and theses to illustrate the points being suggested;
- tasks for you to do which involve observations and analysis of the dissertation and thesis text extracts and reflection on a range of aspects of your own research and literature review.

It can therefore be used as a self-study guide and either read from start to finish at the beginning of the research process to give you a comprehensive view of what is entailed or relevant chapters can be visited as you progress through your research. I would, however, recommend reading Chapters 1 to 5 as you begin the reading for your research because they focus on strategies for conducting efficient searches of the literature, and techniques for reading efficiently and organising the information you collect. Chapters 6 to 10 will become more pertinent when you prepare to write your review and will continue to be relevant as you draft and revise your work. Chapter 11 is a new chapter in this edition and discusses the purposes and processes involved in conducting a systematic literature review. This type of review differs from a thesis or dissertation literature review in that it is a stand-alone investigation usually conducted by a team of experienced researchers. The majority of the book, however, assumes that you are a student conducting research for an undergraduate, masters or doctoral programme or a supervisor of a student undertaking such research.

After working through the guide, I hope that you will have extended your awareness of the issues involved and be able to navigate your way through the maze of identifying and efficiently reading the pertinent literature, and then writing an effective literature review which blends in with your own research.

In this initial chapter, I define the term 'literature review'; discuss the significance of the literature review in research; and suggest alternative ways that the literature review can be positioned in the final draft of a dissertation or thesis, giving the contents pages of dissertation and thesis case studies as examples.

What is a literature review?

When attempting a definition of this complex phenomenon, it is helpful to break it down into two parts: first, the finished *product* of the literature review which appears in the final draft of the dissertation or thesis, and second, the *process* that is involved in conducting a review of the literature.

The 'literature review' is the part of the thesis where there is extensive reference to related research and theory in your field; it is where connections are made between the source texts that you draw on and where you position yourself and your research among these sources. It is your opportunity to engage in a written dialogue with researchers in your area while at the same time showing that you have engaged with, understood and responded to the relevant body of knowledge underpinning your research. The literature review is where you identify the theories and previous research which have influenced your choice of research topic and the methodology you are choosing to adopt. You can use the literature to support your identification of a problem to research and to illustrate that there is a gap in previous research which needs to be filled. The literature review, therefore, serves as the driving force and jumping-off point for your own research investigation.

The term 'literature review' also refers to the process involved in creating the review that appears in your dissertation or thesis. It is an ongoing activity which begins when you pick up the first book or article related to your research and continues until the day you finish the final draft. Initially, your literature review helps you to formulate your research questions; at the same time, you begin to identify the relevant theories and related research studies to your own and the methodology that you might adopt for your research. Later, the literature assists you in the analysis and interpretation of your data. This book explores all aspects of the process including literature search strategies, systems for storing your key sources and notes, reading strategies, and techniques for writing the review.

Some additional definitions of the literature review from other authors are given below to show the varying emphases that different study guides put on its role and purpose.

> Literature reviews should be succinct and … give a picture of the state of knowledge and of major questions in your topic area.
>
> (Bell, 2010: 112)

> [A literature review can be defined as] the selection of available documents (both published and unpublished) on the topic, which contain information, ideas, data and evidence written from a particular standpoint to fulfil certain aims or express certain views on the nature of the topic and how it is to be investigated, and the effective evaluation of these documents in relation to the research being proposed.
>
> (Hart, 1998: 13)

> Typically, the literature review forms an important chapter in the thesis, where its purpose is to provide the background to and justification for the research undertaken.
>
> (Bruce, 1994: 218)

> [The purpose of the literature review] is to locate the research project, to form its context or background, and to provide insights into previous work.
>
> (Blaxter et al., 2010: 124)

> [A literature review should] demonstrate … a fully professional grasp of the background theory to your subject.
>
> (Phillips and Pugh, 2010: 64)

> [In a literature review a writer] extracts and synthesises the main points, issues, findings and research methods which emerge from a critical review of the readings.
>
> (Nunan, 1992: 217)

> [The literature review should be] a coherent argument that leads to the description of a proposed study.
>
> (Rudestam and Newton, 2007: 63)

From these definitions and quotes, we can see that the literature review serves many different purposes and entails a wide variety of activities. It is therefore not surprising that it may seem a formidable task – particularly at the beginning. However, a thoroughly researched and well-written review can be both an exciting and a rewarding experience.

The literature review at different degree levels

Chris Hart (1998: 14–25) summarises the differences in the characteristics of a literature review at different levels of study and outlines the criteria for a successful review. Table 1.1 is taken from his book *Doing a Literature Review: Releasing the Social Science Imagination*.

Jerry Wellington et al. (2005) discuss the professional doctorate and the role of the literature review, in particular in relation to the EdD thesis. They suggest that a literature review for an EdD degree is likely to be more professionally oriented than for a PhD thesis. It is also likely to be shorter as the

TABLE 1.1 Degrees and the nature of the literature review

Degree and research product	*Function and format of the literature review in research at these levels*
BA, BSc, BEd project	Essentially descriptive, topic focused; mostly indicative of main, current sources on topic.
	Analysis is of topic in terms of justification.
MA, MSc, MPh dissertation or thesis	Analytical and summative, covering methodological issues, research techniques and topics. Possibly two literature-based chapters, one on methodological issues, which demonstrates knowledge of the advantages and disadvantages, and another on theoretical issues relevant to the topic/problem.
PhD, DPhil, DLitt thesis	Analytical synthesis, covering all known literature on the problem, including that in other languages.
	High level of conceptual thinking within and across theories. Summative and formative evaluation of previous work on the problem. Depth and breadth of discussion on relevant philosophical traditions and ways in which they relate to the problem.

Source: Hart, 1998: 15

usual length of an EdD thesis is about 50,000 words, in comparison to approximately 70–80,000 words in a PhD thesis.

It is also relevant to mention here that there are other types of literature review which are not undertaken as a stepping stone for a piece of primary research. I'll refer to these two main alternatives as: 1) a stand-alone review and 2) a systematic literature review (see Chapter 11). A stand-alone review is a self-contained literature review on a particular topic which could be conducted for an undergraduate or postgraduate module assignment (e.g. a review on learner autonomy for an education assignment). An entire dissertation or thesis could also be a stand-alone review if it involves an in-depth critical exploration of the literature in a specific area as opposed to being a project including primary data collection and analysis. On the other hand, as mentioned earlier in this chapter, a systematic review is not usually part of a student's assessed degree work. It is a professional and rigorous undertaking which synthesises findings from a sizeable number of individual studies on a topic often with a view to informing a policy or health treatment. Search strategies, article selection, and synthesis and analysis of data are all documented and transparent. I discuss this process in more detail in Chapter 11.

The main focus in this book is on a literature review which leads into a research project for a university degree programme and the final written version appears usually as a chapter or chapters in a dissertation or thesis. Bearing in mind the differences highlighted in Table 1.1, there are some general

principles which can be applied to the process of doing a literature review at all levels and for all degree types, and it is the intention of this book to provide such guidance and suggestions on practice.

Why is it important to undertake a literature review as part of your research?

When you first decide on the problem you think you want to investigate for your research, you may not be thinking extensively about the wider context. But your research is a small piece in a complicated jigsaw puzzle; it does not exist in isolation. It is dependent on what others have done before and you will contribute to an ongoing story or debate. Your reader therefore needs to know about the whole jigsaw puzzle and not simply the shade and shape of your particular piece. In a literature review, you are contextualising your work; you are describing the bigger picture that provides the background and creates the space or gap for your research.

Where do we find the literature review in a dissertation or thesis?

The literature review often appears as a distinctive chapter or a group of chapters in the final draft of the dissertation or thesis. The titles for these chapters vary and can often be topic related instead of being called 'the literature review'. However, it is also possible that the literature review may be integrated throughout the whole thesis and a single chapter is not identifiable. In the latter case, it does not mean that a literature review has not been completed as it is an integral part of all research. The researcher has simply chosen to use the related literature in a more integrated way throughout the thesis. This approach is particularly likely to be the case in research which is based on an analysis of texts and documents as is the case in the fields of history, English literature or biblical studies. Although the extracts from dissertations and theses shown in this book are from identifiable 'literature review' chapters, the guidance offered is also relevant for the integrated literature review as the processes which can be adopted are similar.

How the literature review may be integrated in different theses where there are identifiable literature review chapters and/or sections is illustrated below. There are two main approaches: *dedicated* and *recursive*. In the first approach, the literature review is included in a chapter or series of chapters, frequently with topic-related titles, near the beginning of the thesis. It often begins in the introduction and continues in one or more subsequent chapters. In some fields



Example 1.1 Contextualising your research

A doctoral student in a School of East Asian Studies at a UK university, Key-young Son, was conducting research into President Kim Dae-jung's Sunshine Policy in South Korea. He was investigating strategies of engagement with North Korea and theories of state identity changes which have enabled relations between North and South Korea to improve. He wrote an extensive literature review (the contents page appears below). In a thesis consisting of eight chapters (363 pages), the literature review extended over three chapters and was 144 pages in length (40 per cent of the thesis).

In the first literature review chapter, he explored the political theories of 'containment' and 'engagement' in relation to the Cold War. In the second chapter, a theoretical framework for his thesis was established based on the concept of 'comprehensive engagement'. And in his final literature review chapter, he described the historical context for his study by giving an overview of the changing relationship between North and South Korea. This theoretical and historical overview created a context in which to situate the detailed investigation of President Kim Dae-jung's Sunshine Policy in South Korea between 1998 and 2003.

Key-young shows in detail where his research is positioned within the field of research on International Relations. He does not simply tell his reader about the particular events in the Korean Peninsula's history which relate to the Sunshine Policy. He explains the theories he is using to interpret the events and gives a historical overview of the significant events which have created the context in which the Sunshine Policy was implemented.

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Chapter 8: Conclusion

Source: Son, 2004: ii–vi



Example 1.2 Contextualising your research

The following example is from an MPhil to PhD transfer paper. This is a report which is an integral part of the PhD process for the majority of doctoral students at UK universities. The transfer or upgrade paper (it is known by a variety of different names in different universities) is usually submitted approximately one year after starting on a full-time research degree programme. It typically takes the form of an extended research proposal and outlines the background, aims and research questions, significant and relevant literature, and the proposed methodology of the PhD research. It serves as a milestone when both tutors and the student can reflect on and assess progress and decide whether the student is ready to proceed with the research on a PhD programme.

In this example, Hannah is conducting research in the area of children, food and health. The study has been prompted by a concern with childhood obesity. After a brief introduction where she introduces the topic of obesity and children's eating practices (see contents page below), Hannah includes the literature review in the second section. It is 39 pages in length out of a total of 71 for the whole paper, i.e. it comprises 55 per cent of the report. The literature review is divided into six sub-sections including an introduction and conclusion. The four main content sections discuss and critically analyse literature from a range of disciplines all of which inform Hannah's research on children's perceptions and understandings of health and food.

In the first section, Hannah introduces her theoretical approach to research with children by exploring the literature which argues that the child's voice and perspective is essential. Then, she synthesises research on children as consumers, and identifies a need for further work investigating how children interpret the different messages around them in a range of media and social contexts. After this, she reviews empirical studies which have reported on children's perspectives on food and highlights their limitations; by doing this, Hannah begins to pave the way for her own focus. Finally, the significance of socio-economic position is explored.

Through a critical analysis of the literature under these four headings, Hannah has created a context for her own study and positioned herself as a particular type of researcher in relation to research with children. She has drawn attention to key issues, such as children as consumers and socio-economic position, which will underpin her investigation. She has also emphasised the limitations in previous empirical studies in the field to justify the need for her own.

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Source: Fairbrother, 2010: iii–iv

such as medicine, the complete literature review may appear as part of the introduction under suitable headings and subheadings. Many doctoral theses, in particular, are made up of a number of different research studies which although independent, are connected by the overarching umbrella of the research topic. In these cases, the second approach is more appropriate, with the literature review likely to reappear at various points in the thesis. It begins in the introduction and then continues at the start of each chapter which presents a different study or group of studies. The different areas covered in the literature review for each study are generally referred to in the thesis introduction where the overall background to the research is provided. Examples of both organisational approaches, taken from the contents pages of PhD theses and an MA dissertation, illustrate the various ways in which the literature review can be integrated into the whole thesis (see Examples 1.1–1.7).

Approach 1: Dedicated chapter or chapters to the literature review

In Examples 1.1 above and 1.3 below the literature review appears in a chapter or series of chapters with topic-related titles near the beginning of the

dissertation or thesis. Example 1.2 above and Examples 1.4 and 1.5 below illustrate how there is a dedicated chapter immediately after the introduction which actually has the title 'Literature Review'.



Example 1.3 A dedicated group of chapters to the literature review

Ei leen Lee conducted research in the field of sociolinguistics looking at language shift and language revitalisation among speakers of a specific Creole which is used within a small community in Malaysia. (A creole is a language in its own right which has its origin in extended contact between two or more different language communities. The creole incorporates features of each of the original languages.) Ei leen's literature review begins in Chapter 1, the introduction, with the background historical context of the language, the community where the Creole is spoken, and the national policies which have affected its use. She examines the findings of previous research studies in the field and explains how her research is different from and extends previous research on this particular language. In Chapter 2, she discusses the relevant theories and concepts related to language shift and maintenance which underpin the research and examines three different approaches to the study of language shift in linguistic minority communities. In the third chapter, she explores theories for 'reversing language shift' and considers the role of language planning in the process of revitalisation of endangered languages. These three chapters provide the background which then lead into the description of the methodology, data collection, findings and conclusions.

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Chapter 5: Reported language choice and reported language use at the … settlement

Chapter 6: Language choice and language use in the home and neighbourhood domains of the … settlement

Chapter 7: Language maintenance and competing priorities at the … settlement

Chapter 8: Findings and conclusion

Source: Lee, 2003: iv–vii



Example 1.4 A dedicated single chapter to the literature review

In this example from a PhD thesis in the field of education, Ling Feng explores curricula for education for sustainable development in both China and the UK through three higher education case studies. As well as having a dedicated chapter with the title 'Literature Review', her integration of the literature begins in the Introduction where she clarifies how she is using key terminology for her research: education for sustainable development and the curriculum; she highlights the urgency of sustainable development issues in both the UK and China; and she justifies her research in relation to higher-education institutions, sustainable development and cross national research. For all of these purposes, she cites sources to support her arguments.

In the second chapter, the literature review, she explores the concept of Sustainable Development in much greater depth and gives an international historical overview; the theme of curricula is debated by examining four important curriculum perspectives in relation to Education for Sustainable Development; she discusses the current contexts in Chinese and British Higher Education Institutions with regard to Education for Sustainable Development; and the significance of knowledge sharing between countries. She draws on three of the curriculum perspectives discussed to provide the theoretical underpinnings for the case study research which follows in subsequent chapters.

eg →

Contents

Chapter 1: Introduction

1.1 Background of the research

1.2 The research questions and design

1.3 The justification of the research

1.4 Outline of the thesis

Chapter 2: Literature review

2.1 Introduction

2.2 The notion of sustainable development (SD)

2.2.1 Oxymoron?

2.2.2 Business as usual

2.2.3 Intragenerational equity

2.2.4 Anthropocentrism versus ecocentrism

2.2.5 Complexity and uncertainty

2.2.6 A way forward

2.2.7 Changes occurring

2.3 Historical review of education for sustainable development (ESD)

2.4 Different perspectives on ESD

2.4.1 Technical perspective on ESD

2.4.2 Socially-critical perspective on ESD

2.4.3 Liberal-progressive perspective on ESD

2.4.4 Postmodern perspectives on ESD

2.5 English and Chinese higher education and ESD

2.5.1 English higher education and ESD

2.5.2 Chinese higher education and ESD

2.5.3 Similarities and differences

2.6 Issues of knowledge exchange

2.7. Conclusion

Chapter 3: Methodology

Chapter 4: Research findings

Chapter 5: Discussion

Chapter 6: Conclusion

References

Appendices

Source: Feng, 2010: i–iii



Example 1.5 A dedicated single chapter to the literature review

This example is from a masters dissertation in the field of second language learning. Analeen Moore was researching the influence of formal pronunciation instruction on the English pronunciation of a group of adult German students. Her literature review appears as a single chapter after the introduction and before she begins to describe her experiments in the subsequent chapter. She starts off the review by giving a historical overview of the changes in emphases that pronunciation instruction has received in the field of English as a foreign language (EFL) teaching, and then moves on to define the term 'comfortable intelligibility', which is important when considering target pronunciation. She subsequently examines the effects of instruction, first on second language learning generally and then, more specifically, on the improvement of pronunciation in the target language. In the final part of the review, she defines the features of pronunciation that are analysed in the study when looking at the developments in the pronunciation of the learners in the sample.

eg →

Contents

Introduction
Chapter 1: Literature Review
- 1.1 Historical background
- 1.2 Target pronunciation and intelligibility
- 1.3 Research into the effects of instruction
  - 1.3.1 Second language research
  - 1.3.2 Pronunciation research
- 1.4 Specific features of pronunciation
  - 1.4.1 Strong and weak forms of words
  - 1.4.2 Contractions and elision
  - 1.4.3 Assimilation
  - 1.4.4 Liaison
  - 1.4.5 Stress
  - 1.4.6 Intonation
- 1.5 Summary

Chapter 2: The experiment

Source: Moore, 2001: iii–iv

Approach 2: A recursive literature review

When a thesis consists of a number of different research studies, the literature review is likely to begin in the introduction and then continue at the start of each of the different studies. The reviews before the individual research studies generally elaborate in more detail on the topics raised in the introduction.



Example 1.6 A recursive literature review

This example is from a PhD thesis in the field of psychology for which a series of different studies were conducted to investigate how people self-regulate their behaviour in order to achieve their goals. There were nine research studies in total, which were grouped in four chapters (Chapters 2 to 5). In the introduction, the author begins to discuss previous research and theories related to the regulation of behaviour: goals, motivation and implementation intentions. In subsequent chapters, these areas are revisited in more detail; for example, in Chapter 2 he elaborates on the theories of behavioural intentions and in Chapter 3 more detail is given on goal-achievement models. As before, in the example contents page shown below, a number of the more detailed subheadings have been omitted.

eg →

Contents

Chapter 1: Introduction
- 1.1 The influence of goals on behaviour
  - 1.1.1 Behavioural intentions
  - 1.1.2 Unconscious goal pursuit
- 1.2 The intention-behaviour 'gap' and the problem of volition
- 1.3 Implementation intentions
  - 1.3.1 Accessibility of the specified cue
  - 1.3.2 Strength of the cue-behaviour association
- 1.4 Plan of the thesis

Chapter 2: Meta-analysis of intention-behaviour relations
- 2.1 Behavioural intentions as the cause of behaviour
- 2.2 What other variables influence whether intentions cause behaviour?
- 2.3 Study 1
  - 2.3.1 Method
  - 2.3.2 Results

(Continued)

(Continued)

2.4 Discussion
Chapter 3: Integrating theories of goal-achievement
3.1 Overview
3.2 Models of goal-achievement
3.2.1 Theory of planned behaviour
3.2.2 Goal setting theory
3.2.3 Theory of self-regulation
3.2.4 Strength model of self-control
3.2.5 Emotion
3.2.6 Social support and performance feedback
3.3 Study 2
3.3.1 Method
3.3.2 Results
3.3.3 Discussion
3.4 Study 3
3.4.1 Method
3.4.2 Results
3.4.3 Discussion
3.5 General discussion
Chapter 4: The interplay between motivation and volition
4.1 Overview
4.2 The role of goals in determining the effectiveness of implementation intentions
4.3 Study 4
4.4 Study 5
4.5 General discussion
Chapter 5: The mechanism underlying implementation intentions
5.1 Overview
5.2 Study 6
5.3 Study 7
5.4 Study 8
5.5 Study 9
5.6 General discussion
Chapter 6: Conclusions

Source: Webb, 2003: ii–viii



Example 1.7 A recursive literature review

In this example from the field of civil and structural engineering, the researcher completed three different studies in the area of groundwater contamination, each of which has each been published separately in a peer-reviewed journal. These appear in Chapters 2, 3 and 4 of the PhD thesis. The literature review was visited briefly in the Introduction (Chapter 1) and then elaborated on in further detail in the introductions of the three chapters (Chapters 2, 3 and 4) which describe the different research investigations. In this thesis, the different parts of the literature review are fairly concise and there are no headings and subheadings within the introductions to identify different topics within the literature review. However the adapted contents page below gives an idea of how the review is spread over the three different pieces of research. (Please note, detailed subheadings under 'Method and modelling procedures', and 'Results and discussion' have been omitted from the original in the example below.)

eg →

Contents

Chapter 1: Introduction

1.1 Background

1.2 Objectives

1.3 Scope of work

1.4 Structure of thesis

Chapter 2: Modelling kinetic processes controlling hydrogen and acetate concentrations in an aquifer-derived microcosm

2.1 Introduction

2.2 Method and modelling procedures

2.3 Results and discussion

Chapter 3: Insights gained through multispecies field scale reactive transport modelling driven by biodegradation complexity

3.1 Introduction

3.2 Modelling methods

3.3 Results and discussion

3.4 Conclusions

(Continued)

(Continued)

Chapter 4: Advantages of using adaptive remeshing in parallel for modelling biodegradation in groundwater

- 4.1 Introduction
- 4.2 Model formulation
- 4.3 Test cases
- 4.5 Multi-dimensional field scale simulations
- 4.6 Advantages of parallel processing
- 4.7 Advantages of sparse matrix storage approach
- 4.8 Advantages of adaptive remeshing
- 4.9 Conclusions

Chapter 5: Conclusions

Source: Watson, 2004: 4–6

Structuring your literature review

The above examples of contents pages show how different writers position their literature reviews within the doctoral thesis or masters dissertation and how they break down the related literature into themes or topic areas. As your reading and research progresses, you will find that you gradually begin to categorise your own work in this way, with the final organisational structure and fine-tuning of your literature review being decided by the focus of your research. This process is examined in more detail in Chapter 6.



Task 1.1 Choosing a literature review approach

Ask your supervisor to recommend two or three recently completed dissertations or theses in your discipline. After looking at these examples, answer the following question.

Which of the alternatives for positioning the literature review within a dissertation or thesis, described in this chapter and listed below, seems to be most common in your field?

- The literature review is a distinct chapter or chapters near the beginning of the thesis or dissertation.
- The literature review is introduced in general terms in the introduction and revisited in more detail at the beginning of different research studies in the dissertation or thesis.

- Reference to the related literature is interspersed throughout the whole thesis or dissertation and there is no specific literature review chapter.

Now think about your own research topic.

As a way of helping you choose which alternative may suit your own work, think about the advantages and disadvantages for each alternative in relation to your own topic.

As your work progresses consider which alternative you are most likely to adopt for your own literature review and dissertation or thesis. What are your reasons for making this choice?



Summary

To summarise, this initial chapter has considered:

- the overall definition of a 'literature review';
- a consideration of different types of literature review;
- the literature review's importance in relation to postgraduate research;
- where the literature review may be positioned in the final draft of a doctoral thesis or masters dissertation.

Overview

This introductory chapter has set the scene for the more detailed aspects of the literature review which are examined in the chapters that follow. In Chapter 2, we look further at the multiple purposes of a literature review with extracts from theses and dissertations to demonstrate how the different purposes can be achieved. In Chapters 3, 4 and 5 we consider a range of up-to-date strategies for efficient information searching, evaluating source material, reading and note taking, keeping records, organising source material, and avoiding plagiarism and copyright infringement. Chapter 6 reflects on the writing process and how one can go about organising and structuring the information in a literature review. Referencing conventions are explored in Chapter 7, with coverage of the way references are integrated into the text and how they appear in the bibliography. There is a discussion about the meaning of plagiarism and some examples of the practices that constitute plagiarism. We also consider disciplinary differences in citation patterns and styles. In Chapters 8 and 9, we reflect on what it means to be critical in a

literature review and think about techniques for foregrounding writer voice when integrating sources into one's writing. Chapter 10 discusses the importance of constant revisions and updates of a literature review. We consider the links that are likely to arise between a literature review undertaken at the beginning of the research process and the analysis of the data and discussion of implications arising from your own work. Finally in Chapter 11, the process of conducting and critically appraising a systematic literature review is explored. As part of this discussion, we clarify the differences between a dissertation or thesis literature review and a systematic literature review.

This guide can be used as a reference tool throughout your research, with you visiting particular chapters at the time they are most relevant to you and your work. The chapter aims are clearly stated at the beginning of each chapter to enable you to identify the parts that you need at different stages of your research. Alternatively, you can read the book in its entirety to gain an overview of the whole process. Whichever approach you adopt, the suggestions and examples given should provide you with a basis on which to conduct your own literature review and decide how it can be most effectively integrated into your dissertation or thesis.

2

The multiple purposes of a literature review



Chapter summary

In Chapter 1, a number of different definitions of the literature review were given. In essence, its function is to show how related work in the field shapes and influences your research. The aim is to use the literature selectively and creatively to provide a stimulus for your own work.

This chapter provides:

- a more detailed discussion of the specific purposes of the literature review;
- a selection of extracts from various dissertations and theses to illustrate how these purposes are realised in a cross-section of research contexts;
- a framework to help prompt your thinking about the important elements of your own literature review.

The multiple purposes

When considering the content to include in your literature review, it is important to reflect on the purposes which you wish to address when including references to the work of others. The multiple purposes of the literature



review which appears in your thesis or dissertation can be categorised as follows:

- it provides a historical background for your research;
- it gives an overview of the current context in which your research is situated by referring to contemporary debates, issues and questions in the field;
- it includes a discussion of relevant theories and concepts which underpin your research;
- it introduces relevant terminology and provides definitions to clarify how terms are being used in the context of your own work;
- it describes related research in the field and shows how your work extends or challenges this, or addresses a gap in work in the field;
- it provides supporting evidence for a practical problem or issue which your research is addressing, thereby underlining its significance.

The precise nature of your research will determine which of these purposes receive greater emphasis in your own literature review, but it is important to give each one careful consideration when deciding what your review should encompass.

Some of the functions listed above may be realised in the introduction to a thesis or dissertation instead of in an actual literature review chapter. In Chapter 6, the relationship between the introduction and the literature review is considered in more depth and the purposes which are more likely to be partially or completely fulfilled within the introduction are specified. In a review which is integrated throughout the whole thesis, references to the related literature fulfil the purposes listed above as and when appropriate in the body of the text.

Additionally, in relation to the methodology adopted for your research, you will draw on the literature to justify your choice of a particular approach to the research and the research methods. The references cited can also provide support for the specification and size of the sample you select to be your data, and the sites chosen for your data collection. This particular purpose of the literature is likely to appear in a methodology chapter. Therefore, when we talk about multiple purposes of the literature review, it is important to adopt a flexible approach to the location in the dissertation or thesis where each one is addressed.

As stated in Chapter 1, it is important to show that you have a thorough understanding of the field and that your review of the literature is not a 'laundry list' (Rudestam and Newton, 2007: 62) of previous studies. Essentially, you are building up an argument which leads into your research; and as you develop your argument you draw on source texts selectively to provide support for your assertions. By demonstrating ownership of the argument, you show not only that you are knowledgeable about the field but also that you are entering into a dialogue with other researchers in your field, that is, you are joining a community of researchers in your area. An essential

feature of a successful argument in a literature review is that you make connections between one reference and another, and also explicit links between these sources and your own work. The process of developing your argument in greater detail is also discussed in Chapter 6.

The extracts included below, from a range of academic fields and disciplines, illustrate some of the different purposes that the literature can serve and at the same time show how the writer in each case is building up his or her arguments. You will notice that in many situations, a combination of different purposes can be fulfilled in the same part of the text, although, in my presentation of each example, I have focused on the predominant purpose. The choice of examples is intended to provide snapshots of the different purposes being realised rather than a full picture of each of the literature reviews cited.

Historical background

Your research will almost certainly be situated in a context which has a history. For example, it may be important to describe the sequence of events or policy changes which have led up to the current situation in which your research takes place. Or you might include an account of the development of the research field itself, which documents how knowledge has progressed and perspectives have changed in relation to your topic. As part of the historical background, it may be significant to highlight differing interpretations of events, policies and research findings.



Example 2.1 Historical background

Taken from the field of town and regional planning, this example focuses on the background to contemporary theoretical debates in planning policy. The researcher provides a historical context for the research, situating it against the backdrop of post-war planning and the general stance taken among postmodern theorists.

eg → Disenchantment and disappointment with the results of British post war planning have inspired a 'post-modern' reaction that has sought to reconsider contemporary planning practice, and dominates current theoretical debate. Although those belonging to this post-modern school of thought varied in their approach, most regarded today's planning 'disasters' as the result of an overly authoritarian system which dictated a singular, narrow world view onto society (Harvey 2000). The post-modern theorists were instead concerned for planning to

(Continued)

(Continued)

> realise its full democratic potential by recognising and engaging with difference, recognising that there was not one singular 'public', but many publics with different life-experiences (Healey 1992a, 1992b, 1998; Forester 1989; Campbell and Marshall 2002). There was a concern then with the particularities of individual places, a fascination with local knowledge and cultures (Harvey 2000) and a general foreboding of any universal principles.
>
> Source: Adapted from Coveney, 2003: 5

In a similar way to Example 2.1 above, Example 1.1 in Chapter 1 sets the scene historically for research situated in the post-Cold War period in East Asia. There the researcher creates a historical context for the work in Chapters 2 and 4, entitled 'Containment and Engagement' and 'Historical Overview of the Korean Divide: structure and norms' respectively.



Example 2.2 Historical background

Example 2.2 includes two extracts from a section entitled 'Historical review of education for sustainable development' which is a part of the literature review from the thesis already cited in Example 1.4 in the previous chapter. In this section, Ling reviews the origins of the term, 'education for sustainable development' and also discusses how interpretations and implementations have changed in recent decades.

> **eg →** The evolution of sustainability education has incorporated the significant influence of some of the eighteenth and nineteenth-century thinkers, writers and educators, notably Goethe, Rousseau, Humboldt, Haeckel, Froebel, Dewey and Montessori (Palmer, 1998). Nationally and internationally, Scottish Professor John Smyth is recognised as one of the founders of environmental education (EE), which was both one of the first terms used in relation to sustainability education and one that remains in common usage. He combined his deep interest in nature with his belief in the benefits of education, particularly its ability to bring about change in human thinking and actions, which took him onto the international scene. He participated in writing the first internationally accepted definition of EE which was presented at the International Working Meeting on Environmental Education in the School Curriculum convened by the International Union for Conservation of Nature (IUCN) and UNESCO in 1970 ... (IUCN 1970).
>
> Source: Feng, 2010: 26–7

The term sustainability education officially emerged from the World Summit on Sustainable Development (WSSD) in 2002 as a more inclusive, humanistic version of EE. Having revealed that the goals laid out in the Earth Summit ten years before were still far from becoming reality, the WSSD in Johannesburg identified a clear need to rethink education, and sustainability education paved the way for this 'rethinking' (UNESCO Bangkok, no date). UNESCO (no date) defines sustainability education as 'learning how to make decisions that balance and integrate the long-term future of the economy, the natural environment and the well-being of all communities, near and far, now and in the future' (no page). According to UNESCO (2004), 'the ultimate goal of sustainability education is to empower people with the perspectives, knowledge and skills for helping them live in sustainable societies' (no page).

Source: Feng, 2010: 29



Example 2.3 Historical background

This example comes from the field of health-related research and describes the way that children have traditionally been viewed in society and for research purposes. Later, Hannah goes on to challenge the framework.

eg → Children have traditionally been viewed as objects or 'sociological projects' in the study of child health (Christensen 2004; Mayall 1998). Adult or 'adultist' perspectives have dominated research agendas with three main consequences (Christensen 2004) ... This view of children and its consequences for the research agenda in child health reflected a broader picture within what has been termed the 'dominant framework' for understanding children (James and Prout 1997, p. 10).

The 'dominant framework' has its roots both in sociology and developmental psychology. Working from a sociological perspective, Parsons (1956) highlighted children's need to be socialised to gain an awareness of cultural values and conventions, an empty vessel waiting to be filled with knowledge from adults. Piaget (1955), using a cognitive-developmental theory of growing up, viewed children as lacking in the necessary rationality to make sense of the world. Lee (2001, p. 42) notes:

> The dominant framework endorses the treatment of children as a special case of humanity. It portrays children as peculiarly malleable and ties this malleability to an incompleteness that stems from their proximity to 'nature' and their lack of self-possession.

Source: Fairbrother, 2010: 5–6

Contemporary context

The coverage of the contemporary context in which your work is situated is likely to centre on current topics of debate in the field and the direction of recent and ongoing research. This part of your literature review will link quite closely to the discussion of the significance – and therefore the justification – of your own work.



Example 2.4 Contemporary context

In this extract, the researcher provides the reader with an introduction to the topic of groundwater contamination and current research in the area, within the civil and structural engineering field. This contextualisation also serves to highlight the significance of his topic by giving the reader a sense of why it is important to study different ways of reducing groundwater contamination.

eg → Reactive transport modelling of groundwater systems has become an important field of research during recent years (e.g. Zheng and Bennett 1995; Steefel and MacQuarrie 1996; … Mayer et al. 2002; Prommer et al. 2003). One reason for this is that it is a potentially powerful tool in understanding processes which contribute to the cleansing of contaminated groundwaters. At the field scale, the risk arising from groundwater contamination may be reduced by natural processes (natural attenuation) (Carey et al. 2000; Lerner et al. 2000), or by various engineered remediation methods (Page 1997; … Wagner et al. 2002). Biodegradation is a major process contributing to removal of contaminant mass in plumes of groundwater pollution, and at most field sites is a very significant natural attenuation (NA) process (Smets and Pritchard 2003). Contaminant removal by biodegradation is also preferable to contaminant retardation by, for example, sorption processes, since the mass may be transformed to non-toxic products, as against being stored up for possible later release into the flowing groundwater.

Source: Watson, 2004: 12



Example 2.5 Contemporary context

The following extract gives an overview of the current issues and debates in connection with information management in the National Health Service in the UK. It provides an introduction to the research context of information management in the National Health

Service and, at the same time, defines some of the relevant terminology used in the current study.

eg → Information management in the NHS (National Health Service) has been described as 'health informatics'. This concept is applied to support clinical research, decision-making and practice (NHS Executive 1999). Information management in health care is 'central to … all aspects of information handling in patient care and management of health care services' (O'Rourke 1999: 1). Health informatics emphasizes the use of information technology to manage and provide health services (MacDougall and Brittain 1998). This is quite different to the wider definitions of information management that I have mentioned previously (for example, Best 1996; Parsons 1996; Davenport 1997). These holistic definitions of information management put technology on the periphery, emphasize human involvement and take into account the whole information environment.

MacDougall and Brittain (1998) outlined recent developments and trends in health information. They divided health information into four types: scientific, clinical and health services information; patient-generated clinical data; corporate activity management information; and information for patients, carers and the public. The literature regarding information management can be discussed under the headings, strategies, problems, achievements, needs and resources.

Source: Bacigalupo, 2000: 28



Example 2.6 Contemporary contexts

This extract from the field of child health illustrates the current debate around different paradigms for research with children. It follows on from Example 2.3 above in the previous section which presented the 'dominant framework' as the paradigm that had historically been assumed to be the norm for childhood research. Hannah summarises and interprets the 'new paradigm' which challenges the 'dominant framework', before introducing a further critique which highlights limitations and weaknesses of 'new paradigm'.

eg → It is in this context that James and Prout (1997) critiqued the dominant framework and proposed their 'new paradigm' for researching children and childhood. The key aspects of this new approach were summarised in six points: ...

In each point, James and Prout critique what they see as the major weaknesses in the dominant framework. Childhood as a natural, biological phenomenon is contrasted with their view of childhood as a social construction. The universality

(Continued)

(Continued)

> of childhood emphasised in developmental psychology and the socialisation perspective is replaced by recognition that childhoods are diverse. Research into children's relationships is given value and children are not seen in a vacuum. Children are recognised as social actors, actively constructing their own lives rather than as passive vessels to be filled with cultural awareness (Parsons 1956) and rational thought (Piaget 1955) ...
>
> The new paradigm, the critique of the dominant framework, has also been subjected to critical analysis. Lee (2001) argues that by defining children as beings (in response to the dominant framework's focus on children as 'becomings' on a journey towards becoming fully human) the new paradigm is at odds with the ambiguity of modern childhood and indeed adulthood. ... In a later work, Prout (2005) notes that in focusing on childhood as a social construction, the new paradigm ignores the material components of social life ... In a similar vein, James and Prout (1997) admit that the relativism of the social constructionist approach cannot account adequately for the reality of 'political, social and economic maltreatment ventured against children on an international scale' (p. x) ...
>
> Source: Fairbrother, 2010: 7–8

Theoretical underpinnings

In your literature review, it is important to introduce and discuss any theories and concepts which you intend to draw on to provide direction for your research and to help make sense of your data. Your own interpretation of these theories and concepts should be made clear. This could be done by first comparing and contrasting how others have debated and defined the theories and concepts, and then stating your own position.

The terms 'theory' and 'concept' tend to be interpreted and used in a variety of ways in different fields of research. However, generally speaking, a theory can be described as a framework which offers an explanatory device often in the form of categories and relationships. In the scientific sense, it is likely to be a proposition which both explains and predicts relationships between phenomena. In the social sciences this is not so clear-cut as human behaviour is less predictable and uniform than that of materials and substances.

A concept is a word or expression that represents a general or abstract idea which is derived from more specific instances; for example, 'democracy', 'social class' or 'stress'. In other words, a concept is a representation of an idea in a word or phrase. The use of concepts gives us a means of making sense of the world.



Example 2.7 Theoretical underpinnings

In the excerpt below Son introduces the theories of 'containment' and 'engagement' which provide a framework for his interpretation of relations between North and South Korea. He also relates the theories to the historical context of the research topic by referring to US policy after the cold war.

eg → The aim of this chapter is to provide an historical account of various strategies of containment and engagement and elucidate their theoretical grounding in order to uncover parallels and discrepancies between those strategies and identify the key conceptual components of comprehensive engagement, as proposed by this dissertation.

Containment is a by-product of the Cold War in which the United States and other Western states made all-out efforts to contain the ever-multiplying sphere of 'red' stretching across the Eurasian continent. In his seminal book, Strategies of Containment, Gaddis (1982) sought to analyse containment in terms of 'strategies' as a way to elucidate this illusive concept, which had undergone mutations and transformations through successive US administrations. As part of its global strategies of containment, the United States needed to build close alliances with democratic states worldwide, virtually establishing a wall around the Soviet Union and its allies and giving birth to such metaphors as Winston Churchill's 'Iron Curtain'. Even though the United States placed priority on rebuilding Western Europe, it also allocated limited resources to Asia. First, it turned to Japan in an attempt to find a key security partner in the region, while trying to make the war-devastated South Korea a frontline bulwark against communism (Iriye 1977; Ikenberry 2001a). In particular, the United States opened its market to Japanese and South Korean exports to help to reconstruct the two war-torn states, which could counterbalance the expansion of Communist states in East Asia. Nevertheless, the United States did not remain steadfast in its strategies of containment, as it deviated from its original roadmap to introduce strategies of limited engagement, as seen in the case of its engagement with China from the early 1970s onwards. This section will first offer an historical overview of strategies of containment and engagement, followed by a review of Cold War-related theories from the three perspectives of realism, liberalism and constructivism.

Source: Son, 2004: 14–15



Example 2.8 Theoretical underpinnings

In her literature review, Ling discusses four curriculum perspectives, three of which she draws on to provide a theoretical framework for her research. In the extract below, she introduces the liberal-progressive curriculum and contrasts it with the critical approach to the curriculum. In this part of her literature review, Ling relates each curriculum approach to the Chinese context and also to a range of Education for sustainable development curricula.

eg → Liberal-progressive sustainability education originated from the liberal-progressive curriculum perspective. Both the critical and liberal-progressive curriculum perspectives arose as a reaction against the technocratic and bureaucratic perspectives about and approaches to education in the West. Arising in the eighteenth century, however, the liberal-progressive curriculum perspective formed a distinctively different paradigm (Carr, 1998). While the critical curriculum perspective sees education as a socially transformative rather than socially reproductive agent, the liberal-progressive curriculum perspective sees education as being personal and intrinsically valuable in its own right, instead of as a means or agent for achieving certain goals (Peters, 1966). Inspired by the educational ideals espoused in Rousseau's *Émile* (Rousseau, 1762/1979), the liberal-progressive curriculum perspective focuses on 'rational autonomy and individual freedom' (Carr, 1998, p.327), supporting development of free thinking, responsible and morally mature individuals through personal fulfilment and comprehensive development of the mind, intelligence and imagination (Taylor and Richards, 1985). Liberal-progressive educators attach importance to the discovery and facilitation of learners' intuition from an early age, such as 'intuitive sense of justice' and 'innate child wisdom', throughout and beyond the years of formal education (Ashley, 2005, p.191).

Source: Feng, 2010: 44



Example 2.9 Theoretical underpinnings

In the example below, Hannah begins to draw out aspects from one paradigm for research with children that will underpin her own research. She starts off by recognising that the paradigm she is drawing on has itself been challenged (see Example 2.6 above)

eg → Acknowledging these criticisms, some key concepts of the new paradigm are still useful in working out how to view children and engage in research with them. Matthews (2007) contests Thorne's assertion that the new paradigm has become 'middle aged' (Thorne 2006) and identifies three main ways in which it is still useful. First, it recognises children as playing a part in actively constructing their own lives. Second, there is an emphasis on exploring the plurality of childhoods, how different contexts affect children's relationships, rights and responsibilities. Third, the paradigm emphasises the relational structure of childhood and acknowledges that in these relationships adults have more power. Childhood, therefore, is to be understood as a minority category.

Source: Fairbrother, 2010: 7–8



Example 2.10 Theoretical underpinnings

The following extract introduces us to the theory of 'social constructionism', thus telling the reader about the theoretical perspective which informs the analysis of the social issue being investigated. The reader is informed about the philosophical origins of social constructionism, thereby situating the research historically in the field of sociological theory.

eg → An exploration of how social problems have been conceptualised reveals many broader trends within twentieth century sociology (Rubington and Weinberg 1995). Social constructionism has its philosophical roots in phenomenology (e.g. Schutz 1973; Berger and Luckmann 1966; Douglas 1970), ethnomethodology (e.g. Garfinkel 1973), and symbolic interactionism (e.g. Goffman 1959 and the works of Mead and Blumer). Together they call into question the taken-for-granted nature of reality and see the world as a product of human subjectivity: people define, interpret and give meaning to the world through their everyday actions and interactions.

Source: Calcraft, 2004: 19

Definitions and discussion of terminology used in the research

It is probable that you will use terminology in your research which has been defined and used in a variety of ways by different researchers in your field. This

terminology is likely to encompass the labels given to concepts and theories, such as those mentioned in the section above, as well as more concrete nouns, such as 'social worker' or 'diabetes'. It is important that you do not assume a common agreement about the meaning of all the words and phrases that you include. You should therefore introduce your reader to the variety of ways a term has been defined and interpreted and/or make it clear how you will be using the term (see Chapter 8, Example 8.5).



Example 2.11 Definitions and discussion of terminology

Here the researcher discusses the terms 'diglossia' and 'polyglossia', providing relevant terminology and definitions for the research, and showing how different authors have used these terms in relation to particular social contexts.

eg → 'Diglossia' describes the linguistic situation where … functional specialisation involves the appropriacy of using a particular language or variety, often referred to as the High (H) variety or less prestigious Low (L) variety. … Nevertheless in many multilingual societies, the presence and use of intermediate varieties of languages reveal that the dichotomy between the H and L varieties may not exist as rigidly as in the classic diglossic situation just described. In view of this, Platt (1977) extends the term diglossia to polyglossia to refer to the linguistic situations in Singapore and Malaysia where several codes exist in a particular arrangement according to domains. In the same vein, Fasold (1984), taking examples from countries in Africa, emphasises that in multilingual language situations different forms of diglossia may exist, that is, there may be a High variety and several Low varieties existing alongside each other or different levels of H and L forms overlapping.

Source: Lee, 2003: 69



Example 2.12 Definitions of terminology

The researcher here clarifies the difference in meaning between conscious and unconscious goals for behaviour. Although citing definitions from other authors, the writer is also signalling that these are the definitions of the terms that he will be using for his research.

eg → When considering the influence of goals on behaviour it is useful to distinguish between conscious and unconscious goals. Conscious goals are those which are expressed in the form of a behavioural intention – instructions that people give to themselves to perform a certain behaviour (Triandis, 1980). Consequently, a conscious intention is the expression of an internal goal representation. In contrast, if the person is unaware that a goal is active, or unaware of its potential impact on behaviour, then the goal is deemed to be unconscious (Bargh, 1990; Bargh & Gollwitzer, 1994).

Source: Webb, 2003: 2

Signalling a gap in previous research and using this to justify your own



Example 2.13 The need for extension of previous work

It is a common feature of research generally to identify a 'gap' in previous research which your own work is well suited to fill. This general principle is illustrated in Moore's work, which signals the limitations and paucity of earlier research into the effectiveness of pronunciation instruction and shows how the current study is intended to fill the gap identified.

eg → Turning to research into pronunciation instruction and its effectiveness, the number of empirical studies is fewer and: 'Studies examining the effects of formal instruction in pronunciation have yielded inconsistent and even contradictory results.' (Elliott 1995: 531)

A study by Perlmutter (1989) found improved intelligibility on ESL learners during 6 months of instruction with particular emphasis on pronunciation. However, the subjects were all newcomers to the US and so could be expected to show significant improvement regardless of instruction. … Further, instruction favouring prosodic features such as rhythm, intonation and stress seemed to enable students to transfer their learning to spontaneous situations whereas instruction with only segmental content did not produce such ability.

A study by Zborowska (1997, as discussed in Leather 1999) is reported as finding that explicit instruction helped both the learners' perceptual and productive acquisition of specific phonostylistic processes in comparison with a group which received no instruction.

(Continued)

(Continued)

> In a study of the effects of different types of instruction (teacher correction, self-study in a language lab and interactive modification), McDonald et al. (op.cit. 94) reported 'that no single intervention was beneficial to all learners who experienced it. …
>
> The above studies provide us with neither a definitive answer to the question of whether pronunciation instruction does in fact help improve L2 proficiency, nor to which kind of instruction is most effective. … However, more recent studies would seem to suggest that instruction may be beneficial with the importance of learner attitude and concern for pronunciation accuracy also being important factors in the process (Elliott op.cit.). Further studies into this area are needed to provide a stronger basis on which to plan a course of pronunciation. …
>
> The basic question under examination in this study is whether students who receive instruction in specific features of pronunciation perform better than those who do not.

Source: Moore, 2001: 7–17

The significance of a problem for research

In addition to indicating a gap in previous work, it is equally essential to underline the significance of the research you are proposing. A gap in previous research does not in itself provide justification for your own work. You should give good reasons why your study is important and worth doing. This might include a discussion of your motivation for doing the work; for instance in Example 2.13 the writer emphasised that her research was necessary to enable more effective course planning in the area of pronunciation instruction.

If outlining a problem that your research is intended to solve, it is advisable to incorporate citations from the literature in your field which support your claim that it should be investigated.



Example 2.14 Identifying the problem to be investigated

Both a shortage of research in the area of sustainable urban drainage systems (SUDS) and the importance of conducting more research which assesses the level of pollutants in such systems are indicated here. The author provides supporting evidence of the need

to investigate alternatives to traditional urban drainage systems – the overall problem – while also indicating a gap in knowledge about the way that SUDS operate.

eg → Many professionals are beginning to look towards sustainable urban drainage systems (SUDS) as an alternative to traditional urban drainage systems. SUDS attempt to store, treat and dispose of storm water closer to its source and are increasingly being viewed as effective measures to reduce run off volume, improve runoff quality and promote the recharging of groundwater (CIRIA 2000). Although SUDS have been incorporated into drainage designs for a number of years, they have failed to make a large-scale impact. Many believe this is due to the limited knowledge we have about their actual performance in the field (Delectic 1999) and the wide range of pollutant removals which are reported in various studies (Schueler et al. 1992; Young et al. 1996). It is now widely accepted that urban storm water is polluted and has detrimental effects on our waterways. However, few models exist at present which predict the movements of storm runoff and the pollutants held within it whilst travelling through a SUDS design. Such a model will help to assess the levels of pollutants being contained within SUDS and the levels that continue to reach our waterways.

Source: Overton, 2002: 1



Example 2.15 Identifying the problem and justifying the current research

Practical and theoretical reasons why the research is valuable are provided in this example. The researcher highlights the problem of employer–employee relations after radical organisational change and alludes to the scarcity of research in the area, thereby indicating a gap which the current study can fill.

eg → There is both practical and theoretical significance to a research study of this nature. The potential gains that can be realised if the employer–employee relationship is improved within the … Agency represents the practical significance of this study. … Very few organisations take the time to systematically review the state of their workforce in order to improve the employee–employer relationship, yet it is a key aspect of management training at the university and college level (Fry, Hattwick and Stoner 1998).

The theoretical significance of this study is that there has been very little organisational commitment research conducted in the public sector, let alone within an Agency

(Continued)

(Continued)

newly derived from the public sector. The results of this study will build upon the small amount of existing knowledge that helps public sector employers understand what reactions can be expected from employees following a radical structural and administrative change. These results will also facilitate knowledge-based recommendations that can be employed to better meet the expectations of the employees. To my knowledge, there has not been any published research within the field of organisational commitment that has been conducted within a government institution whose role involves a heritage conservation mandate. Therefore, … this research will reveal organisational commitment results for an Agency … that may differ from other public and private sector organizations. … The more information collected and contributed to the organisational commitment body of literature, the better able managers will be to direct people in a way that best suits the objectives of both the organisation and the employee.

Source: Culverson, 2002: 22–4



Example 2.16 Linking the problem with a justification for the research focus

The justification for the research focus on higher education curricula in relation to sustainable development is given in the extract below. In the previous paragraph, Ling has identified the lack of comprehensive provision of education for sustainable development and here she justifies why she targets higher education institutions for her own research.

eg → HEIs have huge potential to play a significant role in contributing to sustainable development (SD) through sustainability education (Wang and Raninger, 2001; Gough and Scott, 2007). As many HEI graduates will move into responsible decision-making positions, it is important that they are able to make judgments and responsible decisions on various sides of SD issues as individuals and as members of society. Furthermore, as Corcoran et al. (2004) point out, HEIs can also benefit from introducing the notion of SD to HE, because sustainability education presents 'an opportunity to make education more problem based, more interdisciplinary and more applied' (p. 8).

Increasing numbers of HEIs have gradually realised their roles and responsibilities in contributing to sustainable development. The 1990s was a period when HEIs showed their commitments to SD intensively. The Talloires Declaration (1990), Halifax Declaration (1991), Swansea Declaration (1993) and Copernicus Charter (1993), which were all participated in by HEIs worldwide, confirmed their roles

and responsibilities in education, research, operations, policy formation, networking and outreach in relation to SD. In terms of education, the Halifax Declaration encouraged HEIs to 'enhance the capacity of the university to teach and practise SD principles, to increase environmental literacy, and to enhance the understanding of environmental ethics among faculty, students and the public at large' (International Association of Universities, 1991, no page).

Source: Feng, 2010: 7



Task 2.1 Reflecting on your own research

Consider your own proposed research topic.

Using Table 2.1 as a template, consider and note down information which might be relevant for you to incorporate in your review, in relation to each of the suggested purposes of the literature review. In the third column note down the significant references you need to refer to in relation to each purpose. It is likely that you will gradually add to the third column (Key references) as your reading and research progresses.

For this task, think about the information that you want to include and not the order in which you will present it. How to develop your argument and an overall structure for your literature review will be considered in Chapter 6.

TABLE 2.1 The multiple purposes of your literature review

My research topic

Research questions

The purposes of the literature review	*Relevant points to include in relation to your own research topic*	*Key references*
Historical background		
Contemporary context		
Theories and concepts		
Relevant terminology		
Previous research and its limitations (the gap, the challenge, and/or extension)		
The signi cance of the issue being researched		
Other		



Summary

To summarise, this chapter has:

- discussed the multiple purposes which are likely to be realised in a literature review;
- looked at examples from dissertations and theses which illustrate the multiple purposes of a literature review in a range of different research contexts;
- provided a framework which can be used as a basis to explore your ideas about the relevant content for your own literature review.

3

Sources of information and conducting searches



Chapter summary

In this chapter, we are going to consider the importance of conducting a thorough and comprehensive literature search as part of the literature review process. More specifically, the chapter aims to:

- define the term 'literature search';
- discuss the reasons for carrying out a literature search;
- identify the different types of information sources which you are likely to access when involved in academic research;
- provide procedures for evaluating online sources;
- consider the variety of electronic tools you can use to search for information;
- outline an effective sequence of steps to take in the literature search process;
- illustrate the function of key words and Boolean logic when searching for information;
- introduce other electronic tools for supporting the literature search process such as RSS feeds and social bookmarking.

What is a literature search?

A literature search is defined by Sarah Gash (1999: 1) as 'a systematic and thorough search of all types of published literature in order to identify as many



items as possible that are relevant to a particular topic'. It is a crucial part of the research process as it forms the basis of your reading and subsequently the written literature review which appears in your dissertation or thesis. It is an ongoing part of your work. In addition to doing a thorough literature search early on in the research process, it is important to continue to look for relevant reading throughout the project to ensure that nothing is missed and to keep up to date with new publications. It is also possible that the focus of your research may change during the course of your work, which makes it even more important to revisit and adapt your searches to cover this new direction in your thinking.

The availability of information electronically has meant that literature searching has become very much more complex than in the past, although the speed and convenience at which sources can be accessed has also increased. Not only is there a phenomenal amount of information available which you can retrieve from your own computer or mobile phone through the Internet, but also there is a wide variety of ways in which you can search for this information. The choices can be quite daunting when first starting on your research which is why most universities offer courses on information management. Doing a course such as this is time well spent and will help you to identify the most suitable search tools for your academic discipline and particular area of work. Even if you do not take a specific information management course, be sure to allow the library to be your friend and become comfortable about approaching the librarians for assistance. They will not be able to conduct your searches for you but they can offer guidance – and there may be subject specialists who can point you in the right direction for your research topic. Even if you are studying at a distance, there is likely to be a help desk you can phone or email for advice.

What are the purposes of a literature search?

In Chapter 2 we discussed the multiple purposes of the written literature review which appears in the thesis or dissertation, i.e. the literature review product. Now we turn to the purposes of the literature review process and, more specifically, the literature search.

There are a number of reasons why it is important to conduct a thorough literature search. In the first instance you need to identify the field and specific context in which your work is situated. At the beginning of your literature search your reading will be exploratory. You may have a hunch about something you wish to investigate but until you have read around the subject you may be uncertain about the focus and therefore the exact background in which you wish to place your work. Your literature search can give you ideas

about the focal point of your research and the wider context in which it will sit. As well as helping you home in on a topic, it can assist you in identifying your approach to the research and the methodology you wish to adopt. More specifically, it will help you identify the type of data you might collect and use, sites of data collection, the sample size and how you might analyse this data.

In addition to reading around the topic as a means of identifying what and how you wish to research, it is essential that you gain a thorough and comprehensive knowledge of the field. Your literature search and the reading are a means of extending your understanding of the key concepts, theories and methodologies in the field. By increasing your awareness, knowledge and understanding of the area, you will be in a better position to make informed choices about the important research related issues identified above.

Another important reason for a rigorous literature search is to find out what others have done in the area so as to avoid duplicating previous work. This enables you to ensure the originality of your work – an essential prerequisite for PhD research. By exploring previous research in the field, you can identify areas where research has not yet been undertaken, how the work of others could be extended, how questions which previous research has left unanswered can be addressed, and how to avoid repeating mistakes that others have made. In other words, you can identify the gap or niche (Swales, 1990) that your research can fill.

A further purpose of your literature search is to identify key people, organisations and texts which are relevant to your research. Relevant organisations might include academic, government and professional bodies. There will be key journals in your field in which peer-reviewed articles are published. You should identify which journals are relevant for you and keep referring to them on a regular basis throughout your research for significant publications. As you continue to trawl through the literature you will begin to notice familiar names and hence identify key researchers in your area of interest. You may find it useful to search for all their published work and keep an eye on their current publications.

It is clear a range of purposes is served by searching the literature. As stated above, it is crucial to include literature searching as an ongoing part of the research process, although your purposes will shift from being exploratory to becoming more focused as your reading progresses.

Sources of information

In this section, we consider the different information sources that are available. When conducting academic research you will not be restricted to one particular type of publication or information source but will want to investigate

the full range of relevant materials. The most common sources are summarised below.

Books

Books are likely to be the first port of call for many new researchers as a means of beginning the exploratory journey in and around the field. Your university library bookshelves are lined with books in printed form and, increasingly, it is possible to access many of them electronically as ebooks. It is now possible to read many books on a portable ebook reader, such as a Kindle, the device produced by the international online retailer, Amazon. Whether accessed in printed or electronic form, there is a variety of types of book you may refer to as part of your work. The following are the most important.

Textbooks contain the underpinning principles, concepts and theories of a field. They often contain information which is generally accepted as being commonly agreed knowledge in a field.

Specialised books may for example report on research completed for a PhD thesis or a specific theoretical area of a field of research. They may include a collection of chapters written by different authors on a particular specialist theme with an overall editor or editorial board that has collected together and reviewed the contributions for the book. For example, papers from a conference may be published as conference proceedings.

Reference books such as dictionaries, encyclopaedias, directories are used to find specific information or definitions. These are now likely to be available in electronic and/or printed format. Electronic versions are often accessible both online and on CD ROM.

Journal articles

Journals contain collections of peer-reviewed academic articles which are written by different researchers or practitioners in a particular field. Each journal has a specific set of purposes and an intended audience, all of which are usually articulated in each issue that is published. The peer review process means that other academics or professionals in the field have reviewed and made recommendations about each article published, therefore ensuring its quality. Journals include the most recent ideas in circulation in a discipline and the majority are now available electronically.

University libraries subscribe to a large number of journals both in printed and electronic format; however, after conducting your literature search you may wish to access an article in a journal to which your library does not subscribe. It is usually possible to access the article through an interlibrary loan

from a central location such as the British Library. It is also worth doing a search for the article using Google Scholar although you may well find that it is not available free of charge if accessed in this way.

Most journals publish between three and six issues in a single year; the issues in one year are collated into a volume. So, for example, you might see the following reference:

Higher Education 20/3: 260–83.

Name of journal Volume/Issue: page numbers.

Published literature reviews of a field

These may be in the form of either a journal article or a book and provide a summary and synthesis of research and/or theory in a field. They are known as systematic reviews and are usually undertaken by a team of academics and professionals to investigate a clearly focused aim or question. Organisations, such as the Cochrane Collaboration in the healthcare field, provide a collection of systematic reviews to inform practice. A published literature review provides a useful introduction and overview of a field, but it is important not to rely solely on these as, when doing your thesis or dissertation literature review, you should demonstrate that you are able to identify and interpret the relevant original sources yourself (see Chapter 11 for more on systematic literature reviews).

'Grey literature'

Other important sources of information are gathered under the umbrella term of 'grey literature'. The term refers to material that is not published commercially and is difficult to obtain through the usual book-selling and bibliographical channels. Typically, grey literature includes: reports, theses and dissertations, conference proceedings, research in progress, leaflets and posters, media reports, patents, letters, and diaries.

Reports Could be written by companies or government organisations and might report, for example, on an investigation into an event or situation, its findings and recommendations.

Theses and dissertations It can be helpful to access the research of other masters and doctoral students by reading their theses and dissertations. Electronic versions are becoming more widely available; for example, EthOS is the British Library's Electronic Thesis Online Service, through which an

increasing number of UK PhD theses can be accessed. Otherwise, hard copies of theses and dissertations need to be requested from the library where they are held through the interlibrary loan facility or from a country's central library (e.g. the British Library) in printed form or as a CD/DVD. Some theses and dissertations may have been published in a more concise form as either a book or a selection of journal articles.

Conference literature Academic conferences produce collections of printed abstracts and often published proceedings. Proceedings consist of a selection of the papers presented at the conference.

Popular media Newspapers, practitioner trade journals and magazines may be relevant for some research topics so that a researcher can find out about contemporary events and interpretations, and reactions from the general public.

Monographs/work-in-progress papers Some research centres may make specialist papers available which are produced for other researchers and professionals in the field.

Specialist literature and primary data sources In some specialist fields, the literature may also include maps, music, diaries, letters, manuscripts of poems and novels, patents and other legal documents.

Web sites

A range of different types of information can be found through web sites and judgments about the quality of the material accessed have to be made on an individual basis. Some examples of the types of web sites that could be used as sources of information are mentioned here.

Professional organisations such as the British Psychological Association (http://www.bps.org.uk/) can provide a wealth of information on publications, conferences and relevant press releases. UK government sites such as that of the Department for Education (http://www.education.gov.uk/) provide access to policy documents, reports and resources. Charities such as Oxfam (http://www.oxfam.org.uk/) give information about world events and appeals and access to development reports and resources. Increasingly, sites such as these incorporate the use of social media such as Flickr, Twitter, Facebook, blogs, podcasts and YouTube videos, all of which are useful for gaining up-to-date information about the current debates and issues in an area of practice.

Wikipedia (http://en.wikipedia.org/wiki/Main_Page) is an online encyclopaedia which anyone can edit. It is not usually deemed to be an academically reliable source of information but it can, however, provide an overview of a field which must then be explored further through other more rigorously reviewed sources of information. If using a Wikipedia entry to provide you with information, it is essential to evaluate the references cited and access

these to evaluate and use for your own work. Many academic institutions do not accept Wikipedia as a reference in your bibliography due to the open access editing of its pages.

Individual academics usually have their own web pages with their professional details and interests as well as lists of their publications. If you have identified a particular author as key for your research, accessing their personal web page can take you to recent publications which may be important for your work.

Evaluating online sources of information

All sources of information should be evaluated carefully before including in your academic writing and it is especially the case for online resources. This is because anybody can put information on the Internet and the material is not necessarily subject to peer-review quality checks. For a list of references in your academic writing, you should be able to include both the author and date for every source; if either is not traceable it is important to consider carefully whether the reference should be used. As a web site does not always display this information clearly, the points that follow should be addressed to assess its suitability as a source to cite in your work.

- Analyse the URL (Uniform Resource Locator) address to assess the site's credibility: the URL address is the address that you see in the browser at the top of the screen when viewing a web site and the suffixes that appear after the organisation name give you important information. The examples below are home pages of a selection of web sites from different organisation types.

.ac and .edu = educational institutions (http://www.shef.ac.uk/)

.org = non governmental and non-profit making organisation (http://www.farmafrica.org.uk/)

.gov = government organisation (http://www.dft.gov.uk/)

.co and .com = commercial organisation (http://www.canon.co.uk/)

The country code (e.g. .uk for the United Kingdom or .my for Malaysia) tells you the country of origin of the site. (http://www.tourismconcern.org.uk/) (http://www.tourism.gov.my/)

When exploring information on the Internet and moving between web sites, a hyperlink from one web site may take you to any page within another site; it is important to click on the home page tab at the top of a web page to orient yourself in relation to the material. Understanding the URL structure

of a web site is also helpful and often there is a 'site map' tab which will show you the structure of the entire site and enable you to find your way more quickly to the information that you need.

- Who or what organisation has produced the site? Do you already know that this author or organisation has an established reputation in your field? If the author is anonymous, it is advisable not to use the site for an academic reference as you have no way of assessing its credibility. If, on the other hand, you can identify the author or organisation but you do not already recognise them as a reliable source, you should make some further investigations to establish whether they are credible in the field. For example, there may have been a link to this site from a source that you trust. The site may have an 'About us' tab from which you can discover more. You can find out about an author's professional background and qualifications by doing an Internet search for the author or organisation name. Search for other publications by this author in library catalogues and bibliographical databases. All of these techniques can help you establish whether the author or organisation is recognised and established in the field.
- When was the site produced or last edited? Can you find a publication date or a 'last updated' entry on the site. Explore the tabs at the top of the site for this information (e.g. there may be an 'About' tab) or look at the bottom of the page. Again, if you do not know when the information was written, it may not be suitable to use. If a publication date is given but the source is old, assess whether this makes it unsuitable for your purpose.
- What type of online resource is it? If you have accessed a form of social media such as a wiki or blog it will not have been peer reviewed and therefore may not be suitable to cite in an academic paper. The use of information from a social media source may be acceptable in certain disciplines and contexts but this should be confirmed with tutors.
- Is the content presented in a logical way with strong enough evidence? Does the author appear to be presenting a balanced argument? Is there a list of references included and do you recognise them as being credible in your field? If there are errors in grammar, punctuation and spelling, it is likely that the source has not been peer reviewed or even checked and hence is not a reputable source to use. The guidance given on critical reading in Chapters 4 and 8 is also applicable when evaluating online sources of information.

Different types of research

Some of the specialist literature accessed may well be the data or the object of study. Research which uses literature as the data for the study is often referred to as 'armchair' or 'library-based' research and is common in fields such as law, history and literature.

Briefly to clarify this distinction in the way the literature may be used, it is helpful to differentiate between two types of research. For many researchers, their data come from conducting interviews, doing experiments, observing behaviour, etc. The focus is on collecting information that has not been recorded

before. This can be referred to as empirical research, where the researcher is actively going out into the field or working in the laboratory to explore and to discover new information. An alternative kind of research occurs when the data set consists of documents or texts. In this second scenario, the literature itself provides the source of data and therefore is the focus of analysis. Of course, there are some research projects which combine both empirical and library-based research. For example, a researcher investigating patent law might combine both an analysis of the relevant legal documents with interviews with legal practitioners who interpret the laws and regulations.

Tools for finding relevant sources

As stated, many of the information sources you require may be available in electronic form; for example, an increasing number of journals are available electronically through your library web page or open access networks (e.g. Directory of Open Access Journals – DOAJ (http://www.doaj.org/). There may be links to a journal article or electronic version of a book from an individual academic's personal web page.

Having summarised the main types of information sources that you are likely to consult, in this section we move on to consider the main tools you can use to initiate your searches for this information. It is important to spend some time becoming aware of and familiar with the search options available to you as the efficient use of these tools is an essential part of the research process. Broadly speaking, the search tools available can be divided into the following categories:

- catalogues;
- bibliographical databases;
- Internet search engines;
- open access databases;
- professional organisation web sites;

In what follows I describe each one of these and give some examples. To help you identify the most suitable databases and gateways to use in your subject area, you can try an alphabetical search for the relevant subject area on your library web page and/or consult your supervisor and colleagues in your department.

Catalogues

Library catalogues (Online Public Access Catalogues – OPACs) Library catalogues or OPACs provide the bibliographical details and locations of all

the publications held within a particular library. Most UK libraries use Dewey Decimal Classification and in the USA the majority employ the Library of Congress Classification. In both systems, books are categorised according to subject with specific shelf numbers allocated.

The catalogue is likely to be a computer database which can be accessed on the university network. Hence, if you have access to a university computer or an Internet connection you do not actually have to be in the library to use it for your searches. Most operate in a similar way, allowing you to conduct key word searches by subject, author or title. It is a good idea to spend some time becoming familiar with how to carry out efficient searches in your library catalogue in order that you make use of its full potential. Ask a librarian for advice if you need to.

Copac National, Academic and Specialist Library Catalogue This is the combined online catalogue of an increasing number of research universities in the UK and Ireland, including the British Library and the National Library of Scotland and the National Library of Wales. It is a freely available catalogue that enables you to conduct wider searches than possible on an individual university library catalogue and locate where resources are held. More libraries, including some specialist libraries and collections are being added all the time. The Copac web page provides up-to-date information on the catalogue; see http://www.copac.ac.uk/.

National catalogues Many countries have their own national catalogues which list all the publications available in the country. Two examples are:

- British Library Integrated Catalogue. This catalogue operates from the British Library, which has a copy of all the publications available in the UK. It can be accessed from the following web page: http://www.bl.uk/catalogues/listings.html. Zetoc (http://zetoc.mimas.ac.uk/index.html) is the British Library's electronic table of contents for a large number of current journals and conference proceedings dating from 1993. Links are provided to aid access to the full texts of these articles and papers. The resource is freely available to members of subscribing educational institutions and the National Health Service in England, Scotland and Northern Ireland. Email alerts and RSS feeds can be set up so that you receive up-to-date information when new articles and journals are added to the database.
- Library of Congress catalog for the US. From this home page you can conduct searches on the Library of Congress catalogue and access catalogues from many other libraries in the world: http://www.loc.gov/z3950/.

BUBL Link BUBL Link is a catalogue of Internet resources. It provides a link to OPACs of other universities around the world as well as to many other Internet resources. You are likely to be able to access this link from your university library web page on: http://www.bubl.ac.uk/ link/types/opacs.htm.

Publishers' and booksellers' catalogues Booksellers such as Amazon (http://www. amazon.co.uk) and publishers such as Sage (http://www.sage. co.uk/) have catalogues which provide lists of all books published. Searches can be made by subject, author and key word. This can be a useful way of searching for the most recently published books in a field.

Bibliographical databases

University libraries have subscriptions to a multitude of different electronic databases and electronic journals. You usually need university username and password details to access these.

A bibliographical database is an organised list of publications in a particular subject area. The publications included vary between different databases but their main focus is usually on journal articles and conference papers. Bibliographical databases of systematic literature reviews are also becoming an increasingly important source of information. On your university library web pages, it will be possible to search for relevant databases for your subject area and generally speaking you will be able to find a description of each one either on the library web page or when you first access the database. When you identify relevant publications on a database, you will retrieve enough information to then be able to find the publication in a library or through electronic means. In many cases, an abstract will also be provided which gives a summary of the work and you will be able to link to an electronic version of the full text. It is important to note that you are more likely to be able to access the full text of journal articles if conducting your searches when logged on through your university account with a personal username and password than if using an open access search route such as Google Scholar. This is because access may depend on your university subscription to the electronic journal concerned.

Originally databases such as these were available in printed form and, depending on your research, you may still need to consult printed collections of abstracts and indexes if searching for older sources. But electronic databases are now the norm either through networked computer services or on CD ROM.

Some examples of databases include: the Web of Knowledge (WoK) for literature in the sciences, social sciences, arts and humanities; MEDLINE for biomedical literature; ProQuest for ebooks, journal articles and dissertations in a wide range of disciplines; the Cochrane database of systematic reviews in the area of health; Index to theses (UK theses); EthOS (the British Library's Electronic Thesis Online Service); and Dissertation abstracts (US and worldwide dissertations and theses). As mentioned, by logging in as a user to your library web page, you will find the range of databases available which relate to your subject area.

As an individual database may not scan all the journals which may be of interest, it is important to consult more than one database in your field. And because different databases tend to vary with their rules for searches and their use of Boolean operators, it is advisable to use the help menu on an individual database to get tips on efficient approaches to adopt when devising key word combinations.

Internet search engines

Search engines are very large databases. They trawl a large number of other databases and web sites to look for information to create their own database of web pages. Search engines are created by robots which crawl through numerous web pages and add them to the database according to the presence of key words within the web documents. Within a particular search engine, you can look for information by entering key words in the 'search' box. A series of web documents is then retrieved. The most popular and plausible results usually appear first on the list and you can select the ones that look most relevant for your purposes. Two of the most widely known search engines are: Google – http://www.google.com/ and Altavista – http://www.altavista.com/.

Different search engines collect information in slightly different ways, so it is a good idea to become familiar with more than one. You can see the variation by trying out the same key word searches in two or more search engines and comparing the results. Use the site-specific help menu and advanced search option on each one to help you develop efficient searching techniques.

As a word of caution and as discussed above, Internet references are not reliable academic sources if you are unable to trace who produced the material and if the purpose of the documentation is not clear. It is therefore important to use web-based information cautiously and evaluate it carefully before citing it in your academic writing.

Google scholar Google scholar (http://scholar.google.com/) is a more specific search engine which, as the name suggests, provides a tool to search for scholarly or academic web page links. You can search by key word/s or author and find an extensive range of peer-reviewed articles, books, reports and information about the number of times and where these items are cited in other publications. Google scholar also provides information on where and how you can obtain the texts you are looking for.

Open access databases: the Open Archive Initiative

A number of open access repositories of information are now available on the Internet. This means that access to the documents included in a database is

available to all and is not dependent on an individual or institution subscription. Two examples are given below:

- Directory of Open Access Journals (DOAJ). This is a database of open access journals from a range of different subject areas. An increasing number of journals are being added to this directory. A user can gain access to the full text of many of the articles in the journals which are listed. The DOAJ is available on http://www.doaj.org/.
- PubMed. This is a publicly available version of Medline which provides information in the field of health and medical sciences. It is made available by the US National Library of Medicine, on http://www.ncbi.nlm.nih.gov/pubmed/.

Professional organisations As mentioned above, there are many professional organisations with web sites which provide a large quantity of useful information, contacts and links which can lead you to relevant and up-to-date sources. For example, the United Kingdom Literacy Association (UKLA) (http://www.ukla.org/) is a registered charity which promotes effective practice in literacy education in a wide range of settings. It includes links to journals, publishers and resources and provides information about contemporary debates and conferences. Through professional organisation web sites such as these, it is possible to discover the key authors for your field of research, thus helping to ensure you have identified all the significant literature.

Some examples from other fields include the Royal College of Nursing (RCN) (http://www.rcn.org.uk/), the Chartered Institute of Public Finance and Accountancy (CIPFA) (http://www.cipfa.org.uk/), the Law Society (http://www.lawsociety.org.uk/home.law) and The Institution of Engineering and Technology (IET) (http://www.theiet.org/). To find out which professional organisations might be important for your research, conduct a key word search for 'professional organisations *your subject area*' on your Internet browser.

The process of conducting a literature search

In this part of the chapter, the actual process of doing a literature search is considered. In the above section, the focus was on the electronic possibilities for ensuring your search is comprehensive and complete. In what follows, when considering the overall process, I emphasise the importance of spending real time in the library alongside your electronic searches, and the value and role of the snowball technique when you follow up references from the bibliographies of books and articles that you read. Serendipity, the fortuitous and unexpected discovery of something useful, is another important aspect of literature searching. You may find a key source when browsing along the shelves in the library, when chatting to a colleague at a conference or after clicking on a different tag on a social bookmarking site.

Get to know the library

Although more and more information is available electronically, the libraries and their printed resources must not be forgotten. Despite the convenience of being able to access the literature you need from your desk or even your mobile phone, you may still enjoy browsing the shelves and looking at the printed volumes. It is a good idea to start your literature search with a visit to your university library. If you are following a distance-learning course, this is more difficult to do but if at all possible it is helpful to have access to a comprehensive university library throughout your research. Go on an organised library tour if possible to find out the details of where different types of information are stored and to familiarise yourself with the systems for the various facilities you may wish to use. For example, if you are investigating a fairly specific and unusual issue, you may need to read documents which are not in your own university library. To access them you will probably have to make use of the interlibrary loan facility and order books or copies of articles from a central resource, such as the British Library Document Supply Centre. Therefore, it is useful to discover whether and/or how you can make use of this system in your particular university library.

Find the shelves in your university library which have the books most closely connected to the topic you wish to research. A general key word search for your subject in the library catalogue will direct you to the shelves you need. Books will help give you an overview and broad understanding of the field, and help you to identify a more specific topic and key words for more detailed and refined searches. Of course, all the books you need may not be in the same location – especially if you are making novel connections between different disciplines. Also remember that libraries usually have separate shelf sequences and locations for oversized books. Once you have identified the various relevant physical locations of books related to your topic, return regularly and browse along these shelves as you may come across unexpected titles that could be useful. At different points in your research, you will have varying priorities in your mind and different titles may strike you as being important at different times. Electronic books or ebooks are widely available through university libraries now, so also make full use of the online catalogue to find out which ones your library stocks and how to access them.

In a similar way, become familiar with where the journals are shelved. Identify the journals which you think are likely to be most relevant for you and look at the new issues regularly. Journal articles cited in books will help lead you to the key journals in your field. You can also identify relevant journal articles electronically through key word searches of bibliographical databases. Most journals are now available online so it is possible to download articles onto your own computer. Furthermore, in recent years, there has

been a proliferation of journals which are *only* available electronically. Email alerts and RSS feeds (see section below) are widely available tools that can be incorporated into your research activities to ensure that you receive notification of updates to web sites and new publications of relevance.

To take advantage of the electronic search facilities now available, familiarise yourself with your university library web pages, which you will be able to access from your university web site. As you explore these pages, you will find they offer a wealth of resources and advice as well as access to catalogues and links that will take you all over the world searching for information (in the virtual sense, of course). If you are studying at a distance, the online potential of libraries will offer you the means to do much of your literature searching from your own computer or other mobile device with Internet access. To access the electronic possibilities of literature searching, you will need the following: a personal computer with a modem; a telephone connection, if possible with broadband and a WiFi connection; an Internet Service Provider (ISP) such as Tiscali; and browser software such as Internet Explorer. Increasingly, smart phones such as BlackBerrys and iPhones are also being used for Internet browsing although the small size of their screens is a disadvantage.

Consult colleagues and your supervisor

Consult your supervisor and colleagues for references that will introduce you to the field but do your own searches as well. Your supervisor may be happy to advise you on the most useful electronic databases to use for your literature searches but he or she may, equally, deliberately leave this for you to discover for yourself. Part of the challenge of doing research is to be independent and this includes making your own decisions about how and where to search for information and what to read. If, however, you feel uncertain about what you are doing and begin to feel lost or overwhelmed, ask your supervisor or a librarian for help.

Some journals publish review articles which give a synthesis of research and theory in a particular area in the form of systematic literature reviews. If you find one relevant to your topic it provides a good resource but it is important to have a look at the original articles yourself as well. Your supervisor may be able to direct you to such an integrative review if there is a relevant and recent one which relates to your intended topic.

Key word searches

When you have a more detailed idea about the topic you wish to research, you can start to identify your key words and conduct searches on the library catalogue

(OPAC), wider catalogues such as Copac or BUBL, electronic databases, and search engines. The identification of key words involves the selection of nouns and adjectives which most accurately describe the content that you are looking for. When devising your key words, it may be helpful to consult dictionaries, thesauri and encyclopaedias to consider all the alternative words and spellings that cover your topic of interest.

Use the key words specified in articles you read to help you devise further key word searches for yourself. But be aware that even when using key words you may miss articles. The authors of a particular article may have chosen key words to describe their work that are different from the ones you have selected for your search, even though the article may be on the theme that you are looking for. It is therefore important to try out a number of different key word alternatives and combinations.

Keep a record of all your key word searches and where you conduct each one to avoid repeating the same searches at a later date. The systems you can use for organising your searches and reading are discussed in Chapter 5, while Chapter 4 discusses how to cope with the quantity of reading and how to be selective about which texts you actually access from your searches. In the section below on social bookmarking, I explain how tagging is an important development which relates to the use of key words.

The snowball technique

As you read around the subject, you are likely to redefine the focus of your research, which may lead to new or revised key word searches. You will begin to recognise familiar authors and cited texts in the bibliographies of the books and articles that you are reading, and will therefore start to look for more specific texts in catalogues and journals. The snowball technique, when you follow up references from the bibliographies of the texts you read, is well used. In the most obvious sense, this involves following up references to previous work but many electronic databases and journals now allow you to track forward citations. If, for example, you are looking at an article published in 2004, you can find the articles published subsequently in which the 2004 article has been cited. In my experience, extending the scope of your reading by the snowball technique tends to be more common than key word searches as the research becomes more focused and the researcher becomes more familiar with the literature in the field.

As the literature search is a process that will go on for the entire period of your research it is important to keep a record of the steps that you work through. Table 3.1 below as part of Task 3.1 is suggested as a useful checklist for you to follow.

The use of key words and Boolean logic

This section of the chapter includes some general guidance on the use of key words. It is possible to make your key words searches more sophisticated by using Boolean logic. This system is named after the English mathematician George Boole (1815–64). It describes the system of symbols and words that can be used to conduct searches by combining key words into search statements. These search statements allow you to be quite specific about the information you are looking for by indicating what you want included and excluded in your search. Different catalogues, databases and search engines have slight variations in the way they use Boolean logic operators so it is worth checking on their help menus or advanced search options for details.

The basic principles of three of the most useful logical operators – AND, OR, and wildcards – are described here. By combining words with AND, a search will be made for documents that contain both these words. Sources that include only one of these words will be ignored. For example, if you enter obesity AND teenagers, the documents retrieved will contain both the words obesity and teenagers. Similarly, by combining words with OR, a search will be made for documents which contain either of these words. The use of OR is helpful when there are two very similar terms and you wish to broaden the search to find texts which use either word: for example, youth OR teenagers. In this case, the documents retrieved will contain either the word youth or the word teenagers.

The third option is to use a wildcard symbol such as an asterisk – *. The wildcard symbol replaces a character or characters in a word. It can be used with a root word so that all variations of the word will be searched for, or within a word so that alternative spellings will be included. Wildcard symbols vary, so check in the relevant help menu to confirm which you should use on a particular database or search engine. Examples using an asterisk wildcard include:

- col*r – allows for different spellings of colour;
- migrat* – allows for different variations which derive from the root: migrate, migrated, migrating, migration.

As noted above, you will be able to conduct the most efficient searches by looking at the advanced search or help menus on the catalogues, databases, subject gateways and search engines you are using. They offer you different ways of inputting your key word searches. For example, if doing an advanced search using the search engines Google or Google Scholar, the AND, OR, and NOT functions appear using the following wording: Find articles with *all* the words; with the *exact phrase*; with *at least one* of the words; *without* the words (see Screenshot 3.1).



SCREENSHOT 3.1 Google Scholar (c) Google Inc. Used with permission.

In essence, there are two aspects of key word searching which are important:

- the identification of all the key words which accurately and succinctly describe what you are looking for;
- the use of these key words and logical operators efficiently in the various catalogues and databases that you employ for your searches.

Keeping up to date: RSS feeds and email alerts

RSS (Really Simple Syndication) feeds or news feeds are a means of channelling new information from selected web pages to a personal news reader or to the 'favourites' collection on your own computer. By setting up RSS feeds, the latest information comes to one location rather than you having to access each journal, web page, blog, etc., individually to find out whether any new information has become available. An RSS feed can be set up from any online location that displays the RSS symbol . You will need to subscribe to the feed on each web page from which you would like to receive updates by clicking on this symbol and following the instructions. Although you can add RSS feeds to the 'favourites' list on your own computer, you are also likely to need a personal newsfeed reader such as Google reader, especially if you like to access updates from different computers and/or your mobile phone. The news reader constantly checks your selected news sites, databases, blogs and journals for new content and 'feeds' this to your reader. RSS feeds are a helpful method for receiving the most recent information and this is valuable for ensuring that your literature review is current. The

downside is the possibility of an overload of information. Hence, RSS feeds should be set up carefully and each one should be scrutinised to assess its relevance for your topic area.

Another technique for receiving updates involves email alerts. You can register to receive email alerts from some publishers' web pages, journals and databases to match the search criteria that you specify. These alerts will arrive in your email inbox.

Social bookmarking

One further tool for literature searching and also for sharing sources with others who have related interests is a social bookmarking site such as CiteULike (http://www.citeulike.org/). Sites such as this one offer a very different way of finding, accessing, recording and retrieving information and also a way of creating and extending research communities so that the process of collecting relevant literature is not such an isolated activity.

Social bookmarking sites have developed from the commonly used bookmarking facility through which we can add useful web sites to the 'favourites' menu on our web browser tool bar. This enables us to access a web site at a later date with a single click on the 'favourite' link we have created. The limitation of traditional bookmarking is that it is tied to a specific computer and often to an individual user. With social bookmarking, your bookmark is placed on a web site which is accessible from multiple locations and is available to other interested participants.

To use social bookmarking to assist in the literature search process, you first need to select a social bookmarking site such as CiteULike; this particular site has been available since 2004 and is specifically designed for researchers to find, store and share electronically available research papers for their field. After joining the site, links to articles of interest can be saved to a personal library. On CiteULike, bibliographical details are added automatically and personal 'tags' or key words can be created for categorisation purposes. Comments and ratings can be made about the papers added.

In terms of searching for literature, 'tagging' is essential as a user can search for all articles with a particular tag, thus finding other related papers and connecting with others with similar interests who have used the same tags. It is possible to see who else has bookmarked the same article and access their libraries to identify other papers.

There are some additional features of CiteULike which can assist in literature searching. First, RSS feeds can be set up so that you receive notification of new postings and second, bibliographical links can be exported to reference management software such as EndNote (see Chapter 5). Privacy settings are

also available and tag clouds show the most commonly used tags in a larger font than those less frequently used.

There are, however, some potential disadvantages to the use of a social bookmarking tool. There is no monitoring of 'tagging' so accuracy and appropriacy are dependent on responsible and careful use by the participants on a site. In addition, a personal library on a social bookmarking site could become yet one more online location to monitor and update. Thus, for your own research, you must make choices about which resources suit your own purposes and preferences best. Social bookmarking offers the potential of an online research community which may or may not suit your own needs and research context.



Task 3.1 Tracking and recording your search

The checklist template provided in Table 3.1 will enable you to track and record your literature search. In particular, it is important to keep a clear record of which key word searches you have conducted and in which catalogues, databases, subject gateways and search engines. As noted, completing the checklist is an ongoing task. If you are in the early stages of your research, it is important to familiarise yourself with the type of record you will need to keep. If you have already started your literature searches, make sure you have adopted a comprehensive record-keeping system, such as indicated in Table 3.1.

Think of a key word search that you wish to do in relation to your topic, e.g. recycling AND urban OR city OR municipal.

Try your key word search in your library catalogue, an electronic database for your subject area, and an Internet search engine such as Google Scholar. Make a record of all the useful 'hits' on Table 3.1 and compare your results for each search.

TABLE 3.1 A record of your literature search – a checklist

The proposed topic of my research

Key word searches have been completed in the following places:

Name of catalogue, database, search engine or social bookmarking site	*Key word searches conducted or tags used*	*Results of search (e.g. articles and books located)*	*Date of search*

The shelf numbers where I am likely to find books relevant for my research:

Some key books which I have identified for my research:

A useful integrative review of previous work in my field:

Key journals which I have identified for my research:

Important authors/researchers in my field:

Useful professional web sites:

Social bookmarking sites/groups:

Blogs/wikis:



Summary

To summarise, this chapter has:

- defined the term 'literature search';
- explored the reasons why a rigorous literature search is important for research;
- considered the different types of information sources that can be consulted;
- summarised points to take into account when evaluating web sites and online resources;
- examined the various electronic tools available for conducting searches;
- outlined a process to work through when conducting a literature search;
- looked at key words and Boolean operators;
- described the purpose of RSS feeds and email alerts;
- outlined the potential of social bookmarking sites;
- provided a checklist to assist you in keeping track of your literature search.

4

Reading and note-taking strategies



Chapter summary

This chapter overviews effective approaches to reading and note taking, more specifically looking at.

- the reasons for reading;
- techniques for efficient reading;
- the adoption of a critical, analytical and evaluative approach to texts;
- strategies for note taking;
- techniques for writing an effective summary;
- ways of observing and noting connections between different source texts.

In particular, the chapter aims to emphasise the need to be a critical reader: reading actively and analytically, making connections with what you already know and evaluating the strengths and weaknesses of an author's research and arguments (also see Chapter 8, on critical writing).

Let us first of all consider the reasons why reading is such an important part of the research process. We have identified some of the reasons already when thinking about the purposes of a literature search in Chapter 3. Incorporating and expanding on these, we can say that reading contributes to the research process and helps you to:



- find out more about the field in which you are working and identify a topic for the research;
- develop your own understanding of the field;
- find out what other research has been done in the area;
- keep up to date with developments in the field;
- place your own research in a context;
- identify theoretical perspectives that you might wish to draw on;
- find support for your views and arguments;
- explore possible research methods to use for your own research;
- pursue your enjoyment of reading around the subject;
- explore different perspectives in the field and position yourself among them;
- make observations about writing conventions in your discipline to develop your own academic writing style.

At first, it is not unusual to feel overwhelmed by the quantity and variety of literature available. However, as your research topic becomes more fine-tuned, you will gradually become more able to recognise the texts that are most relevant to you and hence the material that you need to read thoroughly and in detail. An important way to cope with the large quantity of reading is to adopt efficient approaches and techniques, first in the selection of the most relevant material to read and, second, when actually reading these texts in detail. If you have a clearly defined purpose in your mind when you read a text, it will help you adopt the most appropriate strategies for the circumstances.

At different stages in the research process, we read for different reasons. In the first instance, our reading is *exploratory*; we want to find out more about the topic and explore ideas in a fairly free-ranging way. Later, we have to adopt a more *focused* approach when more precise research questions have been formulated. At this point, you read for more specific information, such as to find evidence and authority to back up your assertions, contrasting definitions on a particular concept you are using, or reasons for the choice of a particular research method. In fact, not only do we read in different ways depending on our purpose at a particular point in time but also it is common to revisit key sources of information as you will find you take different things from a text according to your focus at a specific moment in the research process.

Techniques for reading efficiently

SQ3R

A commonly recommended procedure for reading efficiently is known as SQ3R (Survey, Question, Read, Recall, Review). By following the suggested steps, you are actively encouraged to interact with the text instead of taking in the

information in a passive and unquestioning way (see Box 4.1). It is, of course, a strategy which can be used flexibly according to your individual preference and purpose for reading at a particular time.

Box 4.1 SQ3R

1 **S**urvey the text to ascertain the gist or general idea.
2 **Q**uestion – while surveying the text, think about questions that you would like the text to answer if you decide it is relevant to read in more detail.
3 **R**ead the text carefully if you think it is pertinent for your research.
4 **R**ecall the main points after you have read the text.
5 **R**eview the text to confirm that you have recalled all the main points that are significant for you and your work.

It is helpful, in addition to the SQ3R framework, to consider in more detail the techniques you might use to work through the above steps.

Survey: skim and scan A central purpose of surveying a text is to identify whether it is relevant for you to read in depth. To do this, skim (read quickly) some or all of the parts which signal the main points: the title, the blurb on the back of a book, the contents page, the index, the abstract of a journal article, the introduction and conclusion of a text, the first and last paragraphs of chapters, and the first and last sentences of paragraphs. After skimming, you should be in a good position to decide whether the source is suitable for a more comprehensive reading.

If you are looking for a specific piece of information in a text, you can scan to find what you require. This means reading quickly until you find the precise detail that you need. For example, it could entail looking for key words in the index of a book to find specific page numbers which cover the topic you are searching for, or it might mean casting your eye over the text itself to spot key words which indicate coverage of the information you want. When you find the important point you are looking for, you can read this part more slowly and carefully. An everyday example of scanning is looking for a phone number in telephone directory or checking a train time on the screen at the railway station.

Question If you decide that a text is relevant for you, then it will be necessary to read it more slowly and carefully in order to remember and understand the content. To prepare yourself to read the text interactively, think about what you already know about the topic; if appropriate, personalise the topic by relating it to your own experience, and predict what you think the text is going to tell you by formulating questions that you hope it will answer.

Read and make connections As you read, make connections with other texts you have read and your own knowledge as this will enable you to understand and remember the new information more effectively. If you are finding it difficult to understand a text, it may be useful to go back to reading less complex material such as introductory textbooks on a topic to build up your background knowledge. This helps you to develop a framework within which to fit more complex information that is new to you.

The note-taking techniques suggested in the section below are relevant at this stage and it is important to find an approach which suits you. You may be in a position to annotate a hard copy of the text with key words in the margin and highlighter pens. If this is the case you may or may not decide to take additional notes as well. However, library books have to be returned, in which case note taking is imperative to help you recall the important information. For these occasions it is helpful if you have a tried and tested system which works for you (see following sections on Techniques for note taking and Making connections between different texts).

To aid your understanding of a text, it can be helpful to map out its organisational structure. We tend to have expectations about how a text will be organised which can help us find the information we need quickly. For example, we usually expect an introduction at the beginning which outlines the content of the text and a conclusion at the end which summarises the main points. There are likely to be minor changes in topic with each new paragraph and more major changes between sections. Section headings indicate the content to follow. In a scientific text, one might expect the following sequence of information: introduction, problem, hypotheses, methods and experiments, results, discussion and conclusion. When taking notes, it can be useful to make a diagram of the structure of the text to illustrate visually how the information connects together.

If you are finding it difficult to digest the content of a text and realise you have to keep rereading pages to extract the meaning, you may find it helpful to break it down into more manageable chunks. The length of a 'manageable chunk' will vary according to the density and language difficulty of the material. It might be a paragraph, a section or a chapter.

Recall and review After each manageable chunk, recall and review your reading by writing a sentence summarising what you have just read. This is likely to ensure you understand and remember the content of the text. Effective summarisation is discussed in detail (see Techniques for writing a summary).

Critical reading

In addition to ascertaining the main points that an author is trying to convey, it is important to adopt a questioning and analytical approach towards the

material. To read critically and analytically, try to answer the following questions in relation to all the texts you use for your research.

- What is the author's central argument or main point, i.e. what does the author want you, the reader, to accept?
- What conclusions does the author reach?
- What evidence does the author put forward in support of his or her arguments and conclusions?
- Do you think the evidence is strong enough to support the arguments and conclusions, i.e. is the evidence relevant and wide reaching enough?
- Does the author make any unstated assumptions about shared beliefs with the readers?
- Can these assumptions be challenged?
- What is the background context in which the text was written? Does the cultural and historical context have an effect on the author's assumptions, the content and the way it has been presented?

Increasing your reading speed

Because of the quantity of reading you will have to do for your research, it is important to read as quickly as is comfortable for you without jeopardising your comprehension of the material. A very important way to speed up your reading overall is to adopt the efficient strategies outlined above so that you are reading in the right way for your purposes at a particular time: i.e. you skim when deciding if a text is relevant for you and read more slowly and carefully, taking notes or annotating, when reading a text in detail. But having said this, if you want to increase your speed generally, the following tips may be helpful:

- Avoid saying the words aloud to yourself as you read.
- Do not run your finger under the lines as you read them; instead, run your finger down the left hand margin of the text to encourage quicker movement of your eyes down the page.
- Endeavour to focus on groups of words as you read each line and not on one word at a time because your natural eye movements involve a series of jumps across the page, pausing to take in each group of words.
- Try not to back track over the text more than necessary. However, a certain amount of regression is a normal part of the reading process as we often look back to check our understanding.
- Attempt to guess the meanings of words you are not sure about. You can look them up in the dictionary later if they are important for your comprehension of the text.
- Vary your speed according to the importance of the information, i.e. slow down for important explanations and speed up for supporting details and examples.
- To increase your reading speed overall, practise as often as possible under timed conditions and complete comprehension questions and/or write summaries to check that you are maintaining a thorough understanding of the content.

Reasons for note taking

When taking notes, you may find it helpful to keep the following reasons for doing so in mind:

- to identify and understand the main points of a text;
- to aid recall;
- to use in later research and writing;
- to aid concentration;
- to make connections between different sources;
- to rearrange information for writing;
- to avoid plagiarism.

Avoiding plagiarism

To expand briefly on the final point above, there are a number of different ways in which plagiarism can occur but the type being referred to here is the use of a cited author's exact words with no or little attempt at rephrasing. Unless these words are part of a direct quotation, this is plagiarism even if the source author is acknowledged in the correct way. Although it may be suitable, on occasions, to use direct quotations when the exact words of a source text are used, their excessive use suggests that you are hiding behind the ideas of the quoted authors and that you do not fully understand their work. Therefore it is essential that you are able rephrase and summarise the points you are taking from source material.

The reasons for and the conventions of referencing, and the various meanings of plagiarism, are discussed in more detail in Chapter 7. Plagiarism is a practice which must be avoided because, in whatever way it occurs, it is considered to be 'literary theft' and suggests a lack of respect for the work of others. When plagiarism occurs, it can result in work being failed or even expulsion from a course of study. It is important to acknowledge the work of others in a respectful way following established conventions.

The suggestions for effective note taking and summarising in the next section provide tips which will assist you in using your own words and hence avoiding plagiarism in your writing.

Techniques for note taking

Features of effective notes

Effective notes are ones that are easy for you to understand at a later time. They generally make use of: key words, e.g. nouns, verbs, adjectives to represent

important points; abbreviations and symbols; a system which shows clearly the difference between main and subsidiary points (this will help you see how the author's ideas all fit together), and a note of all your references.

Box 4.2

Note taking – abbreviations and symbols

e.g.	for example
i.e.	in other words
cf.	compare with
n.b.	note well (important)
>	more than
<	less than
=	equal to
→	leads to

Box 4.3

Example of hierarchical note-taking system

1.

a.

b.

c.

(i)

(ii)

Whichever system of note taking you use, it is vitally important that you record all the details of the reference that you are citing. Make a note of every detail that you will need for the bibliography so that you do not have to search for any of the information at a later stage. If you are including direct quotations, page numbers are essential. In Chapter 5, some of the

choices available for keeping records and organising the information that you read are discussed. In addition, more details are provided on writing bibliographies.

Three main formats for note taking

Annotating a hard copy of the text If you have your own copy of a text, use a high-lighter pen to mark significant points. Write key words in the margins to signal important items. You can use different coloured high-lighter pens for different types of information: for example, you could use one colour for the main idea of a paragraph and another for key examples or supporting information. A third colour could be used for your own comments on the material in the text. Use Post-it stickers to mark important pages in a book.

Pattern notes These take the form of a spider diagram or a mind map, where you put the central idea in the middle of the page and branch out from this with the main themes and sub-themes of the text that you are reading (see Figure 4.1). Mind mapping software such as Mind Genius (http://www.mindgenius.com/) is available if you prefer to do this type of activity electronically. MindGenius has a facility whereby you can set up prompt questions, such as those suggested for critical reading, and address these as you create your mind map.

Linear notes Use headings and subheadings to distinguish between the main ideas and the subsidiary information in a text (see Example 4.1 below).

Your own comments

In whatever form you take your notes, it is important to include your own comments; however, it is also essential to distinguish clearly between the ideas from the text and your own observations. Use a different coloured pen, or font colour if using the computer, to write down your own comments and thoughts about what you are reading: your opinions and questions about the text; connections with your research or other texts you have read; ideas inspired by what you are reading.

It is beneficial to write both a summary of a text and then a short critical review (based on responses to the questions in the section on critical reading above). This will help you recall the content of the text, making it easier to locate the relevant pieces of information and references that you need when you begin to write your literature review.

Another electronic tool that can be helpful for recording information is a digital notebook such as *OneNote*. This is a free-form note-taking facility available as

part of Microsoft Office (http://office.microsoft.com/en-us/onenote/). It is a flexible tool allowing you to store text, pictures, drawings, and audio and video recordings; these can be arranged as you wish on the page. There is great potential for using this tool to make notes on your reading, incorporating your own spoken reactions and drawings as well as more conventional pattern or linear notes. Another similar piece of software that my students have recommended is *Evernote* (http://www.evernote.com/). It was launched in 2008 and has a free version which can be downloaded onto your computer or smart phone.

Handwritten notes vs computer notes

There are some advantages to using a computer system for your notes, whether this is in a specifically tailored package such as EndNote or Reference Manager (see Chapter 5), with mind mapping software, or through the creation of your own system using digital notebooks (OneNote; Evernote), word processing or database files. First, on a computer package there are search facilities which enable you to look for key words within your notes and locate relevant information quickly. Second, you can copy and paste details such as the bibliographical information easily. Third, you can move information around quickly, re-categorising ideas and making new connections.

There are also some dangers, however. With the copy and paste function it is easy to miss the redrafting and rewording stages which are essential for developing your ideas and avoiding plagiarism. If you are using a program such as EndNote you do need to invest time into getting to know the package and if you want all your notes to be on the computer you will need a laptop or smart phone to have with you at all times to accommodate all the occasions on which you may wish to take notes. However, the advantages of storing information electronically are considerable so it is well worth giving some thought to the way you can use the computer and the relevant software to the best effect for your purposes.

But notwithstanding all the new technologies available, it is still invaluable to carry round a small notebook at all times to jot down the ideas that occur to you, for example, when suddenly waking at 3 a.m. or when waiting at a bus stop. These thoughts can then be incorporated into your more systematic note-taking records when appropriate.

Making connections between different texts: using key words

Matching key words that you have devised to identify main points in different texts can help you make links between topics mentioned in the various

sources of information that you read. Often, you will notice contradictions or differences in the literature in relation to research findings or the way theories and terminology are interpreted and used. By writing key words in the margins of your source texts or as headings in your notes, you will be able to find your way quickly to the references or parts of a text where a particular topic is discussed. Observing these connections between different references and making them explicit in your writing is all part of being a critical researcher. In Chapter 6, I show how these connections become relevant when deciding on the structure of your literature review and selecting the different sources that you will use to support the various steps in the argument that you develop in your review.

It is therefore important to work out a system for cross-referencing between the key words identified in your notes on different texts so that you can effectively integrate ideas from different sources in your writing. Again you face the choice between a computer or paper-based system.

A system of note cards is recommended by some (Bell, 2010: 71; Blaxter et al., 2010: 121; Walliman, 2011: 66). If using cards, I would recommend one set of cards for bibliographical details of each reference (see Chapter 5) and another for notes of key points taken from each reference. For the latter set of cards, you can identify the text that the notes are from by writing the author's name and date of publication at the top of the card. As a means of facilitating the shift from reading to writing, if you record one piece of information on each note card, with a clear reference of where it has come from, you can file this information under key words to make connections between texts. When preparing to write, you can lay all your cards out on the floor and shift around the order of the information you want to include. This is a useful way of working out the structure and organisation of your literature review.

You may feel, however, that a system of note cards seems quite laborious and time consuming when computer packages are available with quick and efficient key word search facilities. Nevertheless, your available floor space is probably bigger than your computer screen and therefore physically spreading out and moving around your note cards may give you more flexibility when organising your ideas for your writing. It may be that you adopt a computer based note-taking system at first when you are reading widely and exploring ideas and then consider putting the most relevant points onto note cards when formulating your arguments for your written review (see Chapter 6).

A software package such as NVivo (http://www.qsrinternational.com/products_nvivo.aspx) which is principally known as a tool for the collation, coding and analysis of qualitative data can be invaluable for making connections between different source texts. It offers more sophisticated analysis

techniques than those available on reference manager applications, such as EndNote. This application allows you to analyse the literature in a similar way to the analysis of qualitative data by adding codes to your notes. Within NVivo you can run 'queries' to search for themes across your sources, thus helping you make connections between key words across your references and note the similarities and differences between the ideas of authors being cited. You can either import bibliographical information and your notes about your literature from a reference manager tool such as EndNote (see Chapter 5) or you can take notes directly into NVivo for each source that you read.

Making connections between texts: a tabular comparison

A technique commonly recommended by supervisors to help you achieve an overview of the sources that you are reading is the creation of a table in which you record key information from a group of articles. Recording information in this way enables you to make quick comparisons between studies identifying similarities and differences. The table can provide a helpful stepping stone for the writing especially when reviewing a number of empirical research studies on a similar theme, e.g. investigations into the uses of students' first language in a second language learning classroom. Extracting key information in this way from a source encourages you to be concise and focus on the essential information for comparing and contrasting across references. Sample headings for a table are shown below.

TABLE 4.1 A tabular comparison of sources

Articles on the theme of:					
Author/s (year)	Aims or research questions	Location of study	Sample size and identity	Data collection methods	Key findings

However, prior to a tabular comparison, it is strongly recommended that you take notes in a more detailed manner on each source text either on paper, or using a mind map, digital notebook, reference management software, or NVivo. This ensures you have recorded more detail if needed later on as you progress with your research.

Techniques for writing a summary

Reading is closely connected to the writing process. This connection starts when you annotate, take notes and write summaries of the texts that you read. The close relationship continues while you draft and redraft your dissertation or thesis.

A summary is a concise synopsis of a text which includes the main points of the original text as opposed to all the details. It is different from a paraphrase. When you write a paraphrase, all the information in a sentence or group of sentences from the original cited source is reworded. The aim is to rephrase everything and not to select the most important points. Although there may be occasions when you wish to paraphrase, it is more common to summarise when taking information from source texts as usually you will wish to be selective about the information you incorporate into your own writing.

There are different types of summary and the type you write will vary according to your purpose at a particular point.

A global summary

You may write a global summary when you summarise the entire contents of a text. Global summaries of the key texts you identify for your research are a good idea and provide a useful step between note taking and the selection and integration of source material into your writing. They help you identify what is important. By writing a summary, you become very familiar with the material so that you can recall significant points that you may want to come back to at a later stage in your research and when you are writing your literature review. It is an effective way of ensuring that you understand and remember what you read. It is also a helpful strategy for identifying where there are connections to be made between different source texts.

Abstracts of journal articles, dissertations and theses are global summaries which are created to introduce a reader to the content of the text which follows.

A selective summary

Selective summaries are written when you need to extract some relevant information from a small part of a text for a specific purpose. For example, you may be looking for a reference to support your argument that governments should change their policy on the use of renewable energy sources. In this case, you would not need to summarise entire source texts; rather you would use and acknowledge the parts that are significant for your argument.

Techniques

The following process may help you when writing a summary.

- Make a record of the main points of a text either by note taking or by highlighting and annotating.
- Produce the first draft of the summary by first of all acknowledging the source and then writing down the main points that the text is making.
- Redraft the summary to present the main points in the most logical order. This does not have to be in the same order as the original text. Check that you have not included more detail than is necessary for the purposes of your summary.
- Check back to the original text to ensure you have included all the main points that you need.
- Remember that in a summary you are reporting the author's main ideas. If you wish to make your own comments on the work, be careful to signal it clearly with a phrase such as, 'However, other researchers have found that … ' or 'But in my view … '.



Example 4.1 Linear notes, pattern notes and a global summary

The source text below is followed by some examples of note taking to illustrate both linear notes and pattern notes. Finally a global summary of the text is included.

The source extract

> In contemporary U.S. society, many adolescents spend considerable amounts of time in online interactions (Subramanyam, Greenfield, Kraut, & Gross, 2002). The language used on the Internet demonstrates an evolution of discourse (Crystal, 2001), and adolescents are in the midst of that language evolution (Greenfield & Subramanyam, 2003). Sometimes referred to as netspeak, the language of the Internet entails both traditional linguistic forms and adapted ones that include slang and non-standard forms that are sometimes used in offline life. Netspeak is an emergent discourse that is shaped entirely by the creativity of its community (Crystal, 2001). The introduction of acronyms (e.g., 'lol = laugh out loud,' 'brb = be right back'), plays on variations on words (e.g., 'cya = see you', 'latah = later'), graphical icons that represent emotions, called emoticons (e.g., :) or ;-{}) or graphical icons that represent a real person in a virtual context, called avatars, are all examples of language produced by online communicators. This language continues to evolve and remains an important area of study when considering the ways in which Internet users interact and express who they are.

Scholars have actively explored how identity and language are manifested in online interactions. Current research in computer-mediated communication (CMC) environments such as chat rooms, newsgroups, and multi-user domains (MUDs) has revealed interesting trends in the way individual identity is presented, language is used, and interactions have transpired (Calvert, 2002; Crystal, 2001; Greenfield & Subramanyam, 2003; Herring, 2000; Turkle, 1995). Yet, as new Internet applications are created and embraced, CMC studies must continue to strive toward the understanding of online identity, language, and interaction.

Such is the case for weblogs or 'blogs.' Blogs are personal journals or reversed-chronological commentaries written by individuals and made publicly accessible on the web, and they have distinctive technological features that set them apart from other forms of CMC (Herring et al., 2004a; Huffaker, 2004b). These features include: 1) ease-of-use, as users do not need to know HTML or other web programming languages to publish onto the Internet; 2) ways to archive information and knowledge; 3) opportunities for others to comment or provide feedback for each blog post; and 4) links to other 'bloggers' to form online communities.

These features are especially important for constructing online identity. First, the lack of technical expertise needed to create or maintain blogs makes the application more accessible regardless of gender and age. Next, the ability to archive blog posts creates a way to scaffold on previous impressions and expressions; thus, constructing identity can be a continual process for adolescents, and one to which they can refer. Finally, when blog software offers ways to provide feedback or link to other bloggers, this can foster a sense of peer group relationships, another important aspect for the developing adolescent. In short, weblogs represent a new medium for computer-mediated communication and may offer insight into the ways in which adolescents present themselves online, especially in terms of self-expression and peer group relationships, both of which impact the construction of identity.

Source: Huffaker and Calvert, 2005: 2–3



Linear notes

This example illustrates the use of headings and different layers of subheadings with indentations to indicate the main and subsidiary points from the source text above (Huffaker and Calvert, 2005). Key words are used in the headings and subheadings to capture the essence of the source text.

(Continued)

(Continued)

eg →

Adolescent Internet interactions

1 Weblog = blog = personal journal on a publicly available web page
   a) Relatively new forum for internet communictn
   b) Popular amongst teenagers
      (i) Easy to create as no knowl. of prog. lang. req.
      (ii) Poss. to store postings
      (iii) Can post comments on others' blogs
      (iv) Links betwn blogs
2 Studies of online communctn include analysis of:
   a) Language
   b) Identity
   c) Interactions
3 Features of Internet lang. or 'Netspeak' – evolving with use
   a) Use of acronyms
   b) Word plays
   c) Emoticons (graphical icons)
   d) Avatars (picture icons)
4 Blogs provide forum for study of adolescent identity formation through analysis of:
   a) Self-presentation
   b) Interaction with peers

Source: notes taken from Huffaker and Calvert, 2005

Pattern notes

Figure 4.1 gives a more visual representation of the major themes and sub themes of the source text (Huffaker and Calvert, 2005). The main topic of the text is in the centre, with branches connecting to the more detailed information from the text. Connections are also made between some of the subsidiary points to show additional connections made in the source text.

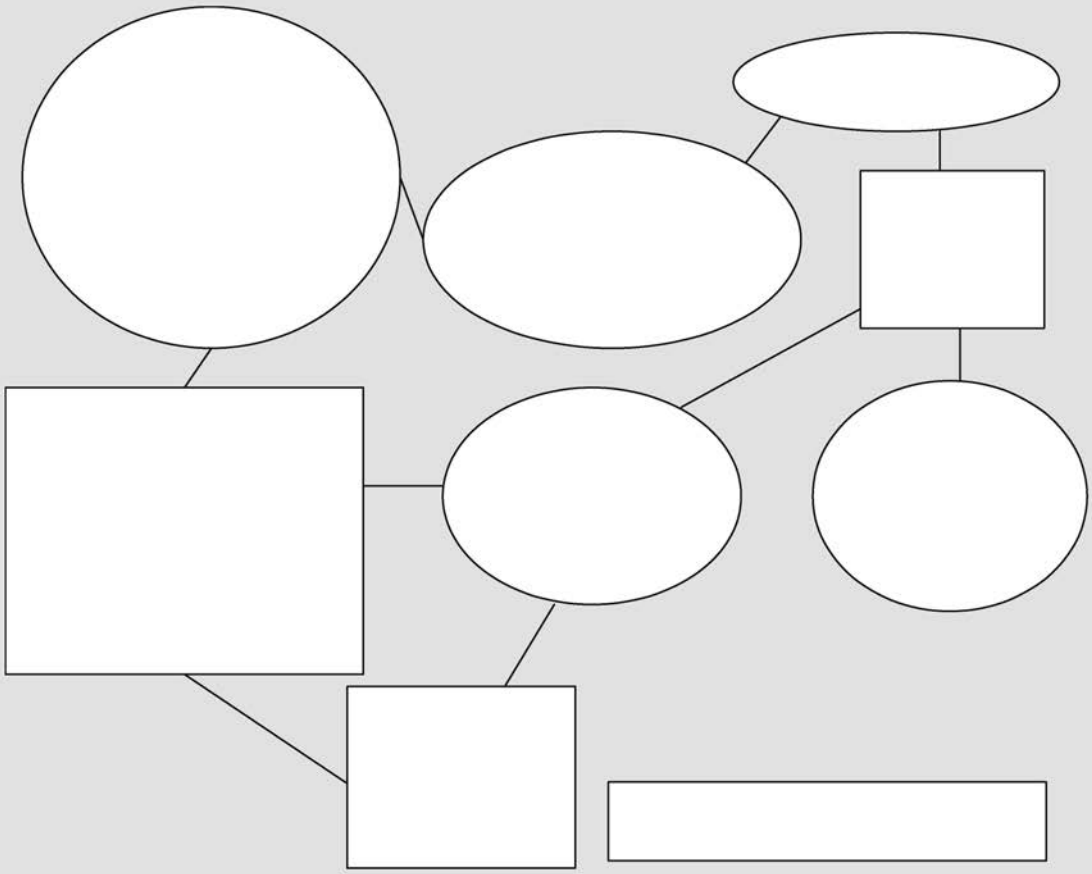


FIGURE 4.1 An example of pattern notes (from Huffaker and Calvert 2005)

A global summary

A global summary which summarises all the key information of a source text aids recall and ensures understanding. The example below gives an overview of the source extract provided above (Huffaker and Calvert, 2005).

eg → Huffaker and Calvert (2005) state that the study of Internet communication has been the focus of much recent research. They elaborate on how this research has included an examination of the construction of identity, the use of language, and the type of interactions that occur online. The focus of some studies has been on the development of Netspeak which is an evolving Internet language that includes icons and abbreviations. A relatively new site for data collection in this area has come about with the development of blogs or weblogs, an innovative type of Internet communication which is particularly popular among teenagers. Blogs are easy to use and require no specialist knowledge of a programming language. They are therefore likely to offer a suitable site for the analysis of adolescent identity formation as the facility for blog postings on both one's own and others' blogs gives opportunities for self presentation and interaction with peers.

(Huffaker and Calvert, 2005)



Task 4.1 Applying the principles to your field

Select and read a research article from your field. Adopt the approach suggested below.

- Survey the text to assess its relevance. Skim the title, abstract, introduction, conclusion and section headings.
- If you decide the article is relevant for your needs, articulate some questions that you would like the text to answer. If the article does not seem relevant for you, try one or two others.
- Read the text more slowly and carefully. If it is your own copy, highlight key points, jot down key words in the margin to remind you of the content and, in a different colour, jot down connections with other texts you have read, or your own opinions and comments about what you are reading.
- You may also wish to take some linear or pattern notes, and/or write summary sentences of each section to remind yourself of the content.
- After finishing the article, look back at the questions you formulated and assess the extent to which the text has answered these.
- Write an overall global summary of the article and then return to the original text to ensure that you have covered all the main points.
- Critically evaluate the article by answering the following questions:
  a What is the author's main argument? What conclusions are drawn?
  b What evidence does the author put forward to support the argument and conclusion?
  c What is your view of this evidence? Can you spot any weaknesses in the methodology, the findings and the way that the conclusions are drawn?
  d Has the author made any implicit assumptions about shared beliefs which you, the reader, you might question?



Summary

To summarise, this chapter has considered:

- why we read;
- how we can read efficiently according to our purpose at a particular time;
- how to read interactively and critically;
- ways of increasing reading speed;
- techniques for note taking and summarising;
- the use of key words to make connections between different sources of information;
- tabular comparisons of source texts.

5

Reference management: keeping records and organising information



Chapter summary

This chapter discusses the importance of and techniques for:

- logging literature searches;
- recording bibliographical details of references;
- becoming familiar with the features of bibliographies;
- developing a filing system for organising notes and hard copies of key source texts;
- avoiding copyright infringement.

In addition, the value and functionality of computer software packages which have specific referencing facilities, such as EndNote, ProCite, RefWorks and Reference Manager, are considered.

Managing the process

In order to manage the literature effectively it is essential to keep records. There are three types of record keeping system which are particularly helpful and you are advised to develop:



- a record of all the key word searches that you do;
- a record of all the bibliographical details you will need for your list of references at the end of your dissertation or thesis;
- a personal library: a filing system of hard copies and/or notes of key texts.

A record of key word searches

As recommended in Chapter 3, it is well worth keeping a log of all the key word searches that you undertake. It is easy to get carried away when doing searches on different databases and catalogues, trying out many variations and different combinations of your choice of key words without pausing to make notes on what you are doing. Without any record of what you have tried out, at a later stage, it will be very difficult to remember exactly where and how you did these searches. It is quite possible that you will duplicate your work and do the same searches again. Thus, a log of all the key word combinations you have tried will enable you to work more efficiently and avoid needless repetition.

Of course, you may intend to repeat your searches if you are looking to see whether there have been any new publications in a field and in this situation as well it is helpful to have a record of the searches conducted previously. By looking back at your records of earlier searches you will be able to revisit databases and catalogues which you have identified as useful in the past and remind yourself of the key word combinations that were successful on prior occasions.

In order for your records to be comprehensive, they should include notes of:

- the key words, phrases and Boolean search terms you used;
- the date that you carried out the search;
- the search engines, databases and catalogues in which you conducted each search;
- the results you obtained from each search.

These records can be kept manually in a notebook, in a Word file or on an electronic database such as Excel (see Table 3.1 for a checklist). You can adopt the format which suits your style of working and preference (or not) for technology. Although it may sometimes seem laborious to keep a log such as this up to date, it can be a real time saver in the long run. It is also interesting to look back over your records at a later date and see how the direction and focus of your search strategy gradually evolved as your research project developed. Electronic applications and databases may have facilities for keeping a record of your key word searches so it is a good idea to use 'help' menus rigorously and explore all tabs to see what is possible.

Additionally, on your notes or hard copies of articles you may find it helpful and interesting to make a note of the path that led you to the reference, e.g. it was found from a bibliography in a particular book, from browsing the new issues of journals in the library, or from a key word search for, say, obesity AND teenagers on the Medline database.

A record of bibliographical details

A bibliography is the list at the end of your dissertation or thesis of books, journal articles, electronic references, and other source texts, which you have referred to in your work. The list may have a variety of different titles, for example bibliography; references; or work cited. Technically, a 'bibliography' can refer to all the source texts that you have referred to while doing your research whereas a list entitled 'references' includes only the references which are actually cited within the main body of the text in the dissertation or thesis. Generally speaking, at postgraduate level, you are expected to show how you have drawn on a particular source text, so in your list (whether entitled bibliography or references) it is common to include only those references which you refer to within the main body of the text. If you have any queries, consult your supervisor to check whether there are particular conventions that you should conform to with regard to the title and the sources which should be included in the list of references at the end of your dissertation or thesis.

As every researcher and librarian will remind you, it is vital to keep a record of all the bibliographical details of every source text that you think you will include in your bibliography. Hours can be wasted searching for an elusive journal name, article title or page number that is needed before the final draft of the work is complete. When there is a submission deadline to meet, it can be especially frustrating to be delayed by missing details for the bibliography.

It is important to familiarise yourself with the referencing conventions of your field so that you can keep a record of all the necessary details from the day you begin reading for your research. Your university web pages will provide details of the referencing expectations of your university and access to other web pages offering further advice. You should also consult any specific guidelines provided by your department as your discipline may have a preferred style and particular recommendations for types of sources that may be specific to your field, e.g. court cases in law or musical manuscripts in music.

Some general conventions which describe the information required in bibliographies for the most common types of information sources are included below.

A book Surname of author, initials of author, year of publication, *title in italics*, place of publication and publisher.

Example: Weart, S.R. (2003) *The Discovery of Global Warming*. Cambridge, Mass.: Harvard University Press.

A chapter in an edited collection or reader Surname of author, initials of author, year of publication, title of chapter (in single quote marks), initials of editor, surname of editor, (ed. or eds), *title of book in italics*, place of publication and publisher.

Example: Scott, M. (1999) 'Agency and subjectivity in student writing', in C. Jones, J. Turner and B. Street (eds) *Students Writing in the University: Cultural and Epistemological Issues*. Amsterdam/Philadelphia: John Benjamins Publishing Company.

A journal article Surname of author, initials of author, year of publication, title of article (in single quote marks), *name of journal in italics,* volume/issue: page numbers.

Example: Shield, F., Enderby, P. and Nancarrow, S. (2006) 'Stakeholder views of the training needs of an interprofessional practitioner who works with older people', *Nurse Education Today,* 26/5: 367–76.

A thesis, dissertation, or project Surname of author, initials of author, year of submission, title of thesis (within single quote marks), level of degree, place of submission.

Example: O'Hanlon, B. (2005) 'Mobile technology for housing association repair departments'. UG final year project, University of Leeds.

Electronic references There are many different types of electronic reference so look for more detailed guidelines on your library web page. Two general principles to keep in mind are: first, you should include enough information for a reader to be able to track down the citation and, second, in order that you cover yourself in case the electronic reference is no longer available, you should include the date that you accessed the information. If possible, all the following information should be included: surname of author, initials of author, year of publication, title of text, URL address, and date accessed.

Example: The Higher Education Academy (2006) Curriculum Design http://www.heacademy.ac.uk/795.htm (accessed: July 22 2006).

As noted in Chapter 3, you must be careful to evaluate the credibility of any electronic reference that you cite. If you cannot identify the individual or organisation who wrote the text, it is probably not a reliable source to be using in an academic context.

The details of punctuation used in bibliography entries vary but the golden rule is to be consistent. One way to decide on the precise format to follow is to consult an important journal in your field and follow their style. All journals have guidelines for authors which describe their exact requirements for the layout of bibliographies. I would recommend that you always have a copy

of the guidelines you are going to follow close to hand until you are familiar with the exact information you need to record, and the punctuation and layout that you are going to adopt.

To record these bibliographical details as you refer to sources of information, you could create a file in Word, use a referencing software package such as EndNote or Reference Manager, or create manual records using note cards (see Chapter 4). EndNote is described in more detail below, showing how it can be helpful when recording the details of different types of references.

If you are likely to include direct quotations in your dissertation or thesis, it is essential that you also record the page numbers in the book or article the quote has been taken from. Ensure you have the exact wording of the quote if you do not have a hard copy of the source text to refer to. The page numbers are not required in the bibliography but they must be included in the text when the quote is used. Conventions for in-text citations are discussed in detail in Chapter 7.

Where you refer to two different publications by the same author/s which were published in the same year, the one mentioned first in the main text should include a lower case 'a' after the year, e.g. (Hyland 2010a) and the second one a 'b', e.g. (Hyland 2010b). The entries in the bibliography should correspond to the in-text references and include 2010a and 2010b after the author's name for each entry, as illustrated below.

> Hyland, K. (2010a) 'Constructing proximity: relating to readers in popular and professional science', *Journal of English for Academic Purposes,* 9/2: 116–27.
>
> Hyland, K. (2010b) 'Community and individuality: performing identity in Applied Linguistics', *Written Communication,* 27/2: 159–88.

Another query frequently raised by students is that of secondary references, i.e. when you cite an author whose work you have not yet seen but who was cited in a source that you *have* read. In this scenario, the in-text reference must make it explicit that the citation is a secondary reference, e.g. (Bhatia 1993, cited in Hyland 2010). The Hyland 2010 reference is included in the bibliography and the Bhatia 1993 secondary citation is not.

The order of the entries in a bibliography can vary. For example, some science and engineering fields may require a numerically ordered bibliography, in which the number for each reference indicates the order in which the references are first cited in the text. However, the majority of disciplines use the Harvard system and will require alphabetical bibliographies where the references are sequenced alphabetically according to the first letter of the author's surname. As described below, EndNote has a facility for creating bibliographies in the style required.

A personal library

The third type of record keeping system which is crucial for your research is a classification scheme for coding and filing hard copies of the texts you collect and identify as being relevant to your research. It is useful to download or photocopy important articles and key chapters from books so that you can refer back to the originals. If this is not possible, make sure you take comprehensive notes and file the notes from these sources. Without a filing system of some kind it will become very difficult to keep track of your references, and as the pile of photocopied articles grows higher and higher on your desk, it will become increasingly difficult to find a text when you want to refer to it again.

To create a coding system, break your topic of research down into subsections and file the literature accordingly in box files or folders. On the cover of the folder, write a name for the category and record quick references to the articles/chapters included inside, either by using author names and/or the article/chapter titles. You can number your folders and use a system of cross-referencing to another folder if a particular reference is relevant in two categories. Although it can become costly, some researchers even like to photocopy multiple copies of key articles so that they can store them in their files for more than one category (but see section on copyright below).

If you have both your own notes and hard copies of articles, you may prefer to develop a separate filing system for each. However, I have found that it is more straightforward to work with a single filing system for both my notes and the original texts so if I take linear or pattern notes of a text and/or write a summary of it, I staple these to my hard copy of an article and file them together. Alternatively, if using EndNote or another electronic application, you can make your notes and write your summary on the computer as part of your electronic entry for each reference. However, you do need to be careful to distinguish between your own notes and summaries and those that you download electronically with the reference. I would strongly recommend using a different coloured font for your own notes if made electronically. You can access your notes quickly by conducting key word searches or opening up the particular reference to which the notes apply. This particular function of EndNote is discussed in a later section.

Copyright legislation

When using source material for your research, it is important to be aware of and avoid contravention of copyright legislation. This is relevant in relation to Internet material, articles and chapters in books that you may wish to download or photocopy for reference whilst conducting your research. It could also be important if you wish to make extensive use of copied maps, illustrations,

tables or similar items, especially if they are from a single source, in your dissertation or thesis.

The Copyright, Designs and Patents Act 1988 provides the basis of copyright legislation in the UK although a number of European copyright directives have also now been incorporated within it. This legislation protects intellectual property when it appears in a physical format, e.g. publications, literary works, computer programs, databases, web pages, art work, sculptures, musical scripts, song lyrics, audio and video recordings, films and broadcasts. The law gives the creator of the work or the copyright owner the right to allow or prevent the copying, reproduction (or performance) of the work and the right to be acknowledged as the author or designer. This right to protection is automatic once the work is committed to paper or another physical form; i.e. registration of the work is not required in the same way as it would be for a patent for an original invention or a trademark.

However, there is an exception to this protection which comes within the scope of 'fair dealing'; it is permissible to make single copies of short extracts (one chapter in a book or one article in a journal issue) of publications as part of research for non-commercial purposes, for educational reasons and for personal use as long as the work and its creator are acknowledged. If accessing an ebook, you will be able to download a prescribed number of pages. Thus, when creating your library of resources for your own research, copyright legislation allows you to copy single copies of extracts of publications within certain limits. In addition, any published work (apart from photographs) may be reproduced for the purposes of review, critique and the reporting of current events. The material must also be acknowledged although this is not a requirement for films, recordings and broadcasts.

There are exceptions to the above rules for those who are visually impaired or have poor hearing. Adaptations can be made to scripts or recordings to enable access for all users of the material.

For more extensive copying of source material, permission must be sought from the copyright owner. For example, in your dissertation or thesis, you might want to reproduce a number of maps or diagrams from a specific source. The copyright owner will be the person who created the work or the publisher of a book or journal. In some circumstances, it may be the creator's employer, for example, if the work has been created as part of the contract of employment. A small fee may need to be paid to the copyright owner for the use of the material and acknowledgement of the work and its creator must be included in the reproduction.

Universities and other educational institutions may purchase licences which can be obtained from an organisation such as the Copyright Licensing Agency to allow multiple copying from specified sources over a period of time. Nevertheless, there are still strict guidelines as to the amount of material and number of copies that can be made. Permission usually includes up to 5 percent

of a work, one article per journal, one chapter per book, one paper from conference proceedings, and one case from a publication of judicial proceedings. The numbers copied should not exceed the number of students on a course and one tutor. Universities may also hold a license to digitise copies of book chapters and journal articles for use with students and licences for recordings and newspaper articles or extracts. Librarians will be able to advise you as to what is permissible under the licences that a particular university holds.

The following web sites provide further information about copyright legislation and should be consulted if you have any concerns about copyright in relation to your own research:

The Copyright Licensing Agency	http://www.cla.co.uk/
Intellectual Property Office	http://www.ipo.gov.uk/

Bibliographical software packages

To assist you in the management of your references, it is a good idea to consider the computer software packages available There is a basic reference management tool available on Microsoft Word (located under the 'References' tab) which may well provide the functionality that you need, for example, if conducting a small-scale research project. This offers the facility to store bibliographical details of the references that you are using, insert citations as you write, and automatically create a bibliography according to your preferred style. However, there are more sophisticated applications available which should also be considered.

I have already referred to EndNote a number of times, indicating some of its functions and when it could be a helpful tool. If you do decide to use reference management software, I would strongly advise that you spend some time early on in the research process exploring its potential and becoming familiar with how to use it. For example, if you are going to store details of all your references in EndNote, it is important to start doing this from the beginning of your research when you first start identifying source texts. If you wait until you are two years into your research, it is a much more arduous procedure to transfer all your records into EndNote libraries.

Bibliographical software packages enable you to:

- fill in a record for each reference using an appropriate and flexible template for each reference type;
- store and organise your references into electronic libraries;
- use the drag and drop function to move references around between libraries;
- input your own notes and summaries for each of your references;
- conduct key word searches for common themes among the references in each of your libraries; searches can be field specific, e.g. according to author, year or key word/s.

- download references and their bibliographical details from electronic catalogues, databases and journals;
- include electronic links to the full text of articles in each reference record;
- conduct searches in remote catalogues and databases;
- include citations from your 'libraries' as you write in Word;
- create bibliographies in your preferred style;
- share your 'libraries' with other researchers.

There are a number of different packages available. The introductory web pages to four important ones are:

- EndNote: http://www.endnote.com/
- Reference Manager: http://www.refman.com/
- ProCite: http://www.procite.com/
- RefWorks http://www.refworks.com/

Your university is likely to support one of the above, thereby enabling you to access the software either free of charge or at a small cost. Look on your library web page or consult staff in the library or computer centre to find out which one is available and how you are able to access it. Alternatively, you can go to one of the above web sites for a demonstration of each package and details of how to purchase it. New versions of each of these bibliographical software packages with additional features are created and launched regularly; for example, mobile applications are now available whereby you can access your bibliographical files when you are on the move through your phone as long as you have Internet access.

As mentioned in the previous chapter, reference manager software can be used in conjunction with the powerful information handling and data analysis tool, NVivo. Bibliographical data can be exported from the reference manager application to NVivo to use the analysis tools for developing and organising your literature review. It can also be helpful to add the bibliographical data to a research project in progress on NVivo. When writing up your project, the references can be imported back into the reference manager application so that you can use the 'write and cite' and bibliography formatting tools which are not available in NVivo.

Using EndNote: the main functions

As an example of a reference manager application, the basic functions and advantages of Endnote are described here. If you do purchase or obtain access to this package, detailed guidelines are provided for its use. But be prepared; the user guide consists of over 600 pages of advice so you will probably wish to refer to it selectively. I outline the main functions of the package and give some tips about how you can start using it. However, you will need to consult

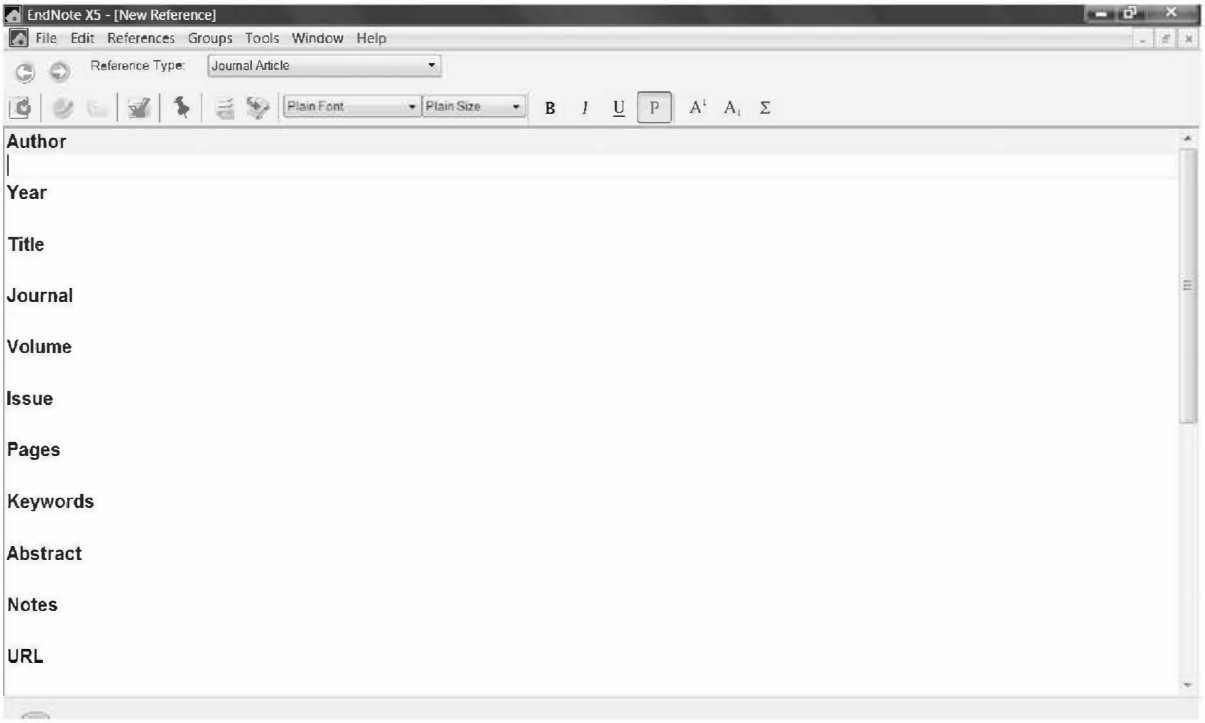
the EndNote user guide or instructions provided by your own university for more detailed help. I would suggest that you select the functions that are useful for your purposes rather than use every facility that the package offers. More recent versions of the package may vary and offer additional features but the basic functionality remains the same. For whichever bibliographical software you decide to use, you will find a 'help' tab within it which will assist you to make a start and offer solutions to questions that arise as you go along.

Reference libraries for storing details of your references Within EndNote you can create database files called 'Reference libraries' in which you can store individual references. To create a new library, select 'Create a new EndNote Library' from the opening window of the Endnote program (see Screenshot 5.1). By creating a selection of libraries with different names you can categorise your reading into topic areas. If this screen does not appear when you open EndNote, go to the *File* menu and select *New* or *Open* to create a new library or open an existing library.



SCREENSHOT 5.1 © 2012 Thomson Reuters

Within the reference libraries you can store all the bibliographical information you need for each reference that you wish to use for your research. For each different reference type (for example, a journal article, a book, government document), EndNote provides a set of fields which are appropriate for this specific reference type. For example, for a journal article, you fill in the following fields: Author/s, Year of publication, Title of article, Name of the journal, Volume, Issue and Page numbers. There are also a number of other fields such as those which enable you to include the abstract of an article, incorporate an electronic link to the article itself, or add your own notes (see Screenshot 5.2).



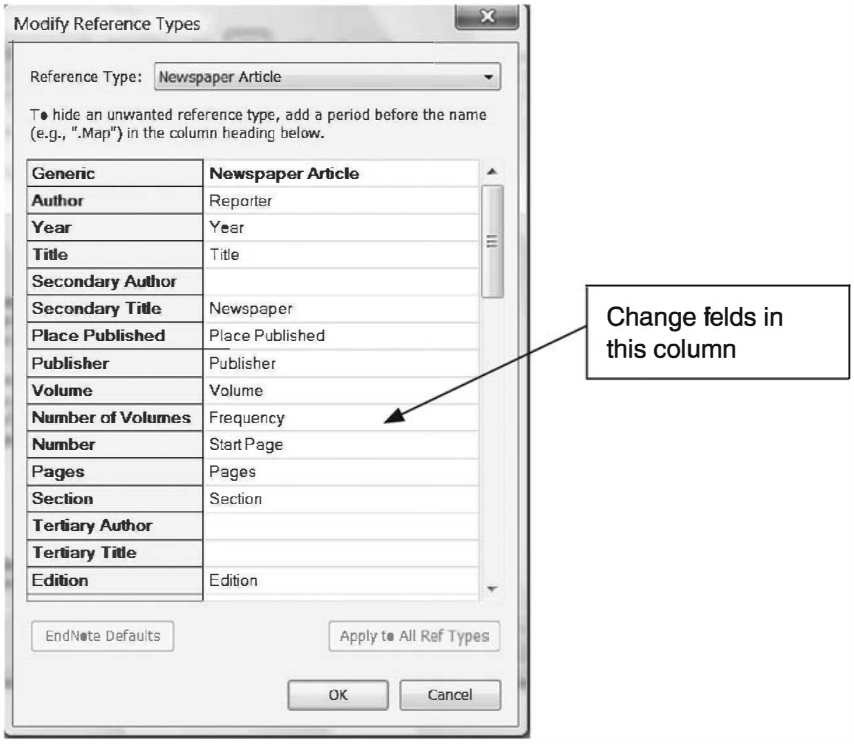
SCREENSHOT 5.2 © 2012 Thomson Reuters

EndNote provides you with the facility to edit the fields available for each reference type:

- Choose 'Preferences' from the 'Edit' menu.
- From the list which appears on the left of the EndNote preferences box, select 'Reference Types'.
- From the drop-down list on the right of the box, choose the Reference Type that you wish to change, e.g. 'Newspaper article'.
- Click on 'Modify Reference Type'.
- In the column on the right in the new box which appears (see Screenshot 5.3), you can delete, rename or add fields.
- You have the option of applying the field change to all reference types which you can do by clicking on the 'Apply to All Ref Types' button.
- Click 'OK' to save your changes.

Key word searches It is possible to locate references stored in Endnote quickly and efficiently through key word searches.

- Open a reference library.
- If the search boxes are not visible in the lower half of the screen, click on 'show Tab Pane' in the bottom right hand corner.
- Select a key word or phrase and specify the fields that you want to search, e.g. the 'Title' field.
- It is possible to use the Boolean search terms, And, Or and Not to make your search either more specific or more wide reaching.



SCREENSHOT 5.3 © 2012 Thomson Reuters

For example, the search shown in Screenshot 5.4 is for any reference which includes within any of its fields, either of the words 'assessment' or 'testing' in conjunction with the term 'higher education'.

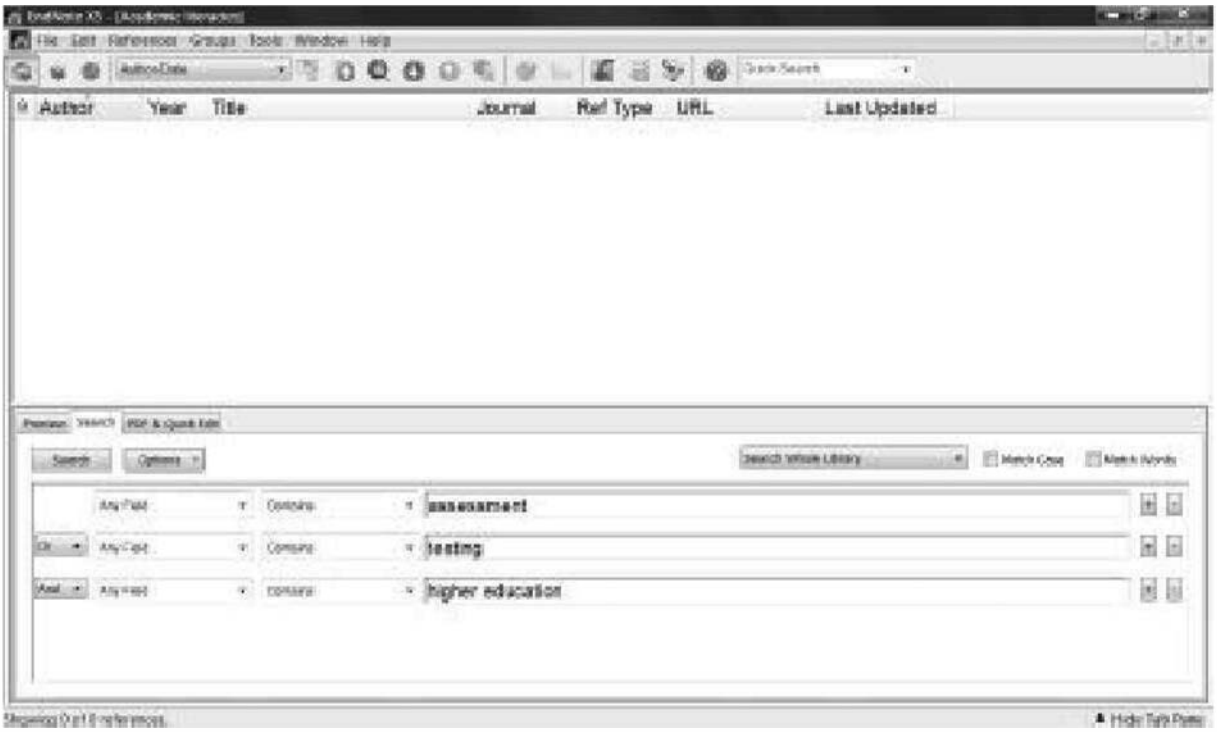
Importing references – manual data entry You can import references into these libraries either manually or by importing them from other electronic bibliographical databases. To enter a new reference manually into a reference library:

- Select 'New reference' from the 'References' menu, or click on the new reference icon on the tool bar.
- Select a 'Reference type', such as, a journal article or a book, from the drop-down menu.
- EndNote offers you fields to fill in which are appropriate for that reference type, as shown in Screenshot 5.2.

You can add your own notes and key words to the bibliographical entries to identify a source text. I would strongly recommend using a different coloured font for your own additions, so that at a later date you can clearly distinguish between your comments and those written by the author of the source text. This confusion is most likely to arise when you import electronic references from electronic journals or databases as the author's abstract and key words are usually automatically inserted into the appropriate fields.

Reference entries can be edited at any time by clicking on the reference in the reference library and moving your cursor to the field you wish to change.

Downloading references from remote databases When accessing electronic databases, catalogues or journals directly through your university



SCREENSHOT 5.4 © 2012 Thomson Reuters

library connections, you can often import the bibliographical details of your selected references directly into your EndNote libraries using EndNote's import filters. Not all databases will offer you this option, but the possibility of downloading information in this way is becoming more widespread all the time. The filters play an important role as, by selecting the right filter for the database or catalogue that you are accessing, you save the data in a way that is compatible with the EndNote fields for each reference type. The difficulty is that every remote database appears to have its own system for downloading references to packages such as EndNote, some of which are more straightforward than others. If you meet any problems, consult help menus and librarians for assistance.

Example 5.1 Downloading an electronic journal article details into your EndNote library

eg → To illustrate the procedure, this example lays out the steps necessary for downloading a reference to EndNote from an electronic journal. This type of process is becoming more and more common as information management becomes

(Continued)

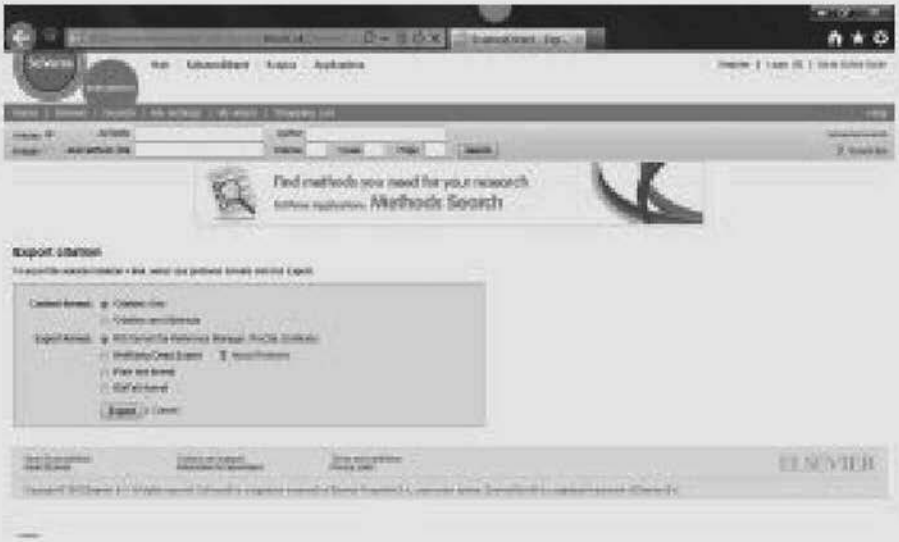
(Continued)

increasingly computer based. Where available, you can download all the bibliographical details that you need, the abstract and an electronic link to the article itself. The procedure generally involves marking the relevant article with a tick in the box beside its title in the contents page of the journal and then selecting the 'Export' button (see Screenshot 5.5).



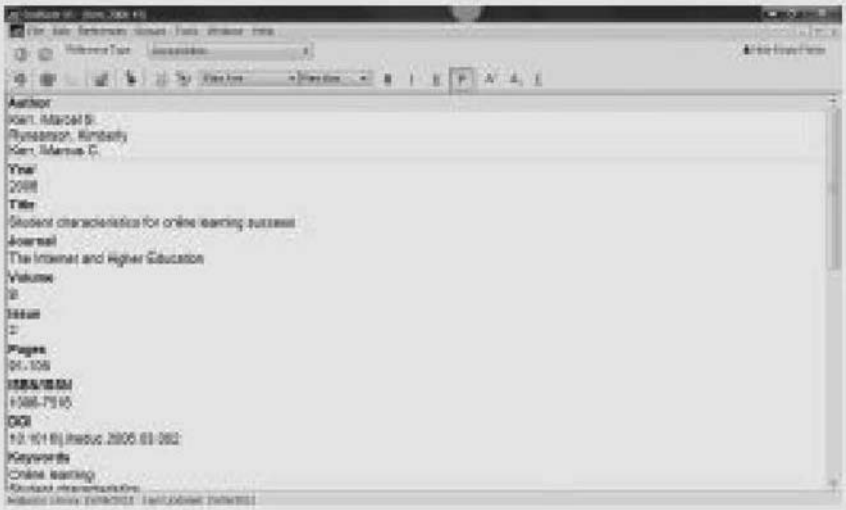
SCREENSHOT 5.5 http://www.sciencedirect.com

You usually then have the option of selecting EndNote which, if loaded onto your computer, will automatically open. You select the reference library into which you want the details of the article to go and the reference downloads automatically (see Screenshot 5.6).



SCREENSHOT 5.6 http://www.sciencedirect.com

Screenshot 5.7 shows an example of the reference as it appears in the EndNote reference library.



SCREENSHOT 5.7 © 2012 Thomson Reuters

Conducting searches in remote catalogues and databases You can also connect to a large number of online catalogues and conduct key word searches from within EndNote and download references into your EndNote libraries. Once connected to a specific online catalogue, such as the British Library, you can carry out a key word search in the same way as you would within EndNote. From the references found in the catalogue, you can download the bibliographical details into your own Endnote libraries. However, you cannot specify the articles that you want to download and you still have to access the source text itself either through an interlibrary loan service or electronically if the article or book is available in this form.

There are some other drawbacks to be aware of when conducting searches of remote databases through EndNote. You do not have access to some of the facilities that would be available if you accessed the database directly. For example, when searching through EndNote you cannot access the advanced search features of the remote database or its online help menu and you do not see links to the full text of articles. Therefore, you are likely to find it preferable to access the electronic database, catalogue or journal directly and then import the references to EndNote.

Cite from EndNote while you write and create your own bibliography You can cite from your EndNote libraries while you write a document in Word. You may need to configure Word to work with EndNote, so consult the computer centre in your university on how to do this. If EndNote is loaded onto your computer and is already configured, 'EndNote' will appear on the toolbar when you are working in Word.

Before starting to use this facility while writing a document in Word, you should select and amend the bibliographical styles to that which you are most likely to use. EndNote has over 1,000 different styles stored which are mainly labelled by journal name. There are also some generic styles offered, such as 'author–date' or 'numbered'.

- In EndNote, from the 'Edit' menu, select 'Output' styles'.
- Choose 'Open Style Manager'.
- From the styles and journal names which appear, select the styles you are most likely to use and these will then be available when you are working both in EndNote and in Word.

If the journal style that you are looking for is not there, choose a style that is similar to what you want and *amend* it. If you are following the Harvard system of referencing, amending the generic *author–date* style with your specific punctuation and font preferences can be an effective way of getting the style that you want. To modify a style:

- From the 'Edit' menu, select 'Output Styles'.
- Choose 'Open Style Manager'.
- Click on the Style info/preview button to see examples of each style in the lower window.
- Change the setting from 'Style info' to 'Style preview'.
- Scroll through the styles and select the one you wish to edit, for example 'author–date', by clicking to place a tick in the box beside it and highlighting the full line.
- Click the 'Edit' button.
- When the 'Style Window' opens, from the 'File' menu, select 'Save as'.
- Choose a name for the new style. This will mean that the original style will remain unchanged for further use if necessary.
- In the left-hand column in the 'Style Window', select the aspect of the citation or the bibliography that you wish to change.
- For example, if you wish to change the way titles of books are written in bibliographies from being underlined to italics, click on 'template' under 'Bibliography'.
- In the window that appears, make the necessary changes.
- Click 'Save' from the 'File' menu and the amended style is available for use.

This procedure can be fiddly and it can take some time to get the citations and bibliography appearing exactly as you want but, once achieved, it can save you a lot of time in the long run.

Before you start writing in Word, it is a good idea to open EndNote and select the library you are going to be referencing from. To select the style that you want to use:

- From the toolbar in Word, select EndNote.
- Choose the style of referencing from the drop down menu that appears.

- Click on the Bibliography menu in the 'Format Bibliography' dialogue box, next to 'Format document', select the Word document you wish to format.
- Next to 'With output style' select the bibliographical style you want. The most common would be 'author–date'. If you have amended this style and given it a new name, select this one and click OK.

As you create a document, when you need to insert a citation:

1 From the 'Tools' menu in Word, select EndNote.
2 Choose 'Find Citation'.
3 In the 'Find' box you can either type a key word, an author surname or a year.
4 Click on 'Search'.
5 A list of items from the reference library which matches the search terms will appear.
6 Scroll down and select the one you need.
7 Click 'Insert'.
8 The reference will be inserted within your text.
9 As you write, EndNote automatically creates a bibliography for you at the end of the document in the style you have specified.

Creating an independent bibliography You can also create an independent bibliography which is not associated with a specific paper. This is likely to be the case when you wish to create the bibliography for your dissertation or thesis. In fact, you may well find this a more useful and straightforward way to create the bibliography that you need than the 'cite while you write' process outlined above. There are several different ways to create an independent bibliography, all of which are described in the EndNote user guide. One technique is outlined below to give an indication of the facility.

- Select the bibliographical style that you want by choosing 'Output Styles' from the 'Edit' menu.
- Select 'Open Style Manager'.
- Scroll down and select the style that you want.
- To do this, highlight the style, ensure the box is ticked beside it, and close the 'Style Manager' window by clicking on the cross in the right-hand corner.
- As above, if you have amended a style for your own personal use, select this one.
- Open the library from which you wish to create your bibliography. You may find it helpful to create a specific reference library which will form the basis of the bibliography for your dissertation or thesis.
- Select all the references that you wish to export to your bibliography by holding down the 'Shift' key.
- From the 'File' menu in EndNote, select 'Export'.

- From the 'Save File as Type' list, choose RTF (Rich Text Format) if you wish to create an editable Word document.
- Enter a file name and select the folder in which you wish to save your references.
- Click 'Save'.



Task 5.1 Record keeping for your own research

In the light of the issues raised in this chapter, consider your own record keeping systems for the literature you will be consulting throughout your research. Use the following checklist to reflect on how you are currently working and consider how you could develop your practice.

Literature record checklist

- ☐ I am keeping a record of all my key word searches.
- ☐ I have identified useful catalogues, databases, journals and authors that I will continue to refer to during my research.
- ☐ I have implemented a system for recording the bibliographical details of all the references I consult.
- ☐ I am aware of and have an example of the bibliographical conventions of my field, both for in-text citations and the bibliography.
- ☐ I am beginning to categorise my reading into topic areas.
- ☐ I have developed a filing system for storing my notes and hard copies of source texts.
- ☐ I have explored the potential of an electronic referencing package supported by my university (e.g. RefWorks, Reference Manager, ProCite, or EndNote).

I am using the above reference management software to do the following:

- ☐ Store references in reference libraries
- ☐ Keep my own notes on references
- ☐ Download electronic references
- ☐ Create my bibliography
- ☐ Other _____



Summary

To summarise, this chapter has looked at ways of managing the references that you use for your research. In particular, we have emphasised the importance of keeping records and developing a filing system for:

- key word searches;
- bibliographical details of all your references;
- your notes and hard copies of the source texts that you use.

An overview was also given of the significance of copyright law and the potential uses of referencing software packages, including guidance on how to implement the following functions in EndNote:

- the creation of electronic reference libraries;
- the manual input of bibliographical details and notes for references into EndNote libraries;
- key word searches within EndNote libraries;
- the download of references from electronic databases, catalogues and journals;
- searches in remote databases and download references;
- the citation of references from EndNote libraries while writing a document in Word;
- the modification of a citation and bibliographical style to suit personal requirements;
- the creation of an independent bibliography from an EndNote reference library.

Acknowledgement

EndNote X5 © 2012 Thomson Reuters.

6

Structuring the literature review



Chapter summary

This chapter considers:

- the cyclical nature of all the activities involved in the creation of a literature review;
- the close relationship between the processes of reading and writing;
- ways of structuring the review;
- the complementary relationship between the introduction and the literature review;
- some examples from theses and dissertations which illustrate how different researchers organise their use of the literature.

The processes involved in the creation of a literature review

Searching for the literature, reading the source material and writing the review are all interconnected and cyclical processes. There is no clear cut-off point when one activity ends and another begins. Indeed, although there may be an intensive focus on the literature review in the earlier weeks and months of a research project, the processes connected with the review continue to be interwoven throughout the research project. In particular, Wellington et al. (2005) emphasise the significance of continually revisiting the research questions or research focus to help you determine and adapt more precisely what and how much you read in relation to your research topic. Figure 6.1 illustrates the continuous, cyclical and interconnected processes which all contribute to the literature review. The literature searching, reading and writing feed into each other constantly; and all the other activities,



such as formulating research questions and justifying the research problem, influence and are influenced by the literature searching and reading, providing inspiration for the writing. Your writing in turn helps you discover and clarify your ideas and can result in the refinement of the focus of your research and the content of your review.

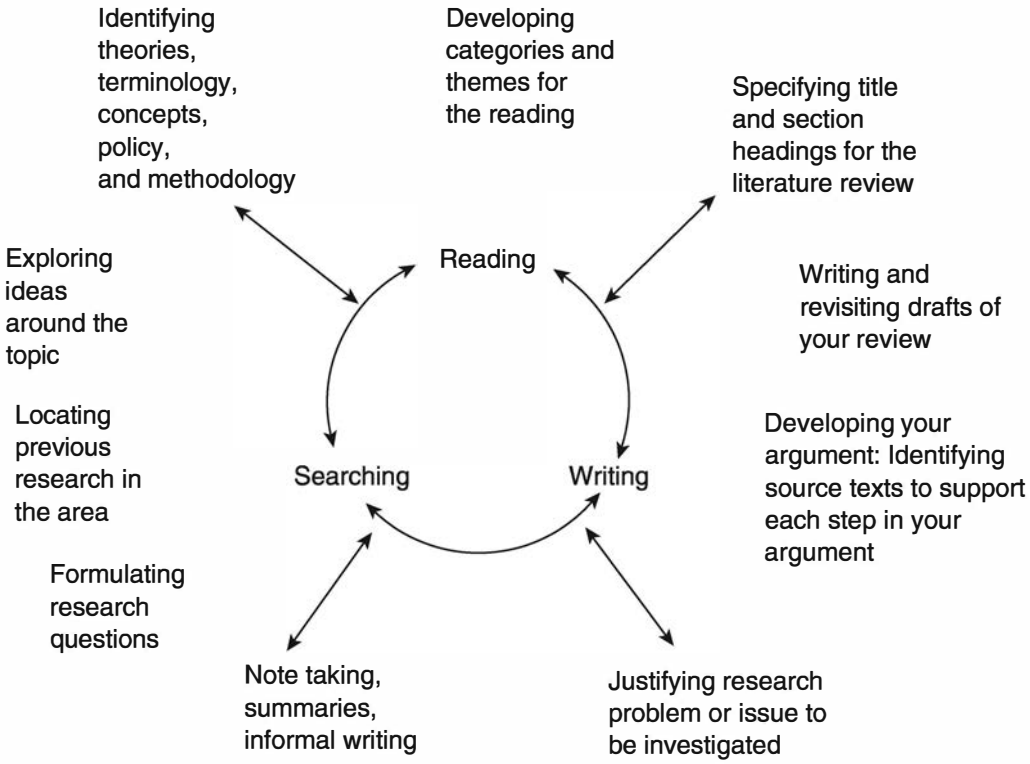


FIGURE 6.1 The literature review process

Beginning to write

It is a good idea to begin writing about the literature as early as possible and it definitely is not necessary to wait until you have a finalised plan or outline. Indeed, starting to write about the literature before the overall organisational structure of your review is clear in your mind is a means of helping you to understand the literature, and discover and clarify how you want to use it (see the sections on note taking and summarising in Chapter 4). In this vein, Murray (2011: 110) suggests various techniques to guide informal writing about the literature before moving on to more formal structured writing. She recommends regular writing to prompt questions, such as: 'What do I know about my research topic?'; 'What I am looking for in the literature is … '; 'What are the schools of thought in the literature?'; 'The "great debates" in my area are … '.

This type of regular writing is a means of establishing what you already know, identifying what further reading you need to do, clarifying how your research links with that of others, and developing your own understanding

and interpretation of the literature. Some of your informal writing may form the basis of drafts for your actual literature review which will appear in your dissertation or thesis.

The structure of the literature review

Gradually, as your reading, note taking, summarising and informal writing progresses, you should try to pull together the various threads and move towards an arrangement with headings and subheadings to provide a framework for your review. The categories that you develop to organise your reading material can often provide the basis for the outline of your written review. You will have your own preference for the stage at which you devise an outline for your literature review; some researchers like to do this very early on in the process whereas others prefer to spend longer exploring the literature in a more free-ranging way. But at whatever point you decide on an outline, it will almost certainly change and evolve through a series of drafts.

Because every piece of research is different, it is difficult to suggest a common organisational structure but some suggestions are given below to help you think about the possibilities. The important point to remember is that in your review you should present a logical argument that leads smoothly into your own research, justifying both the need for work and the methodology that is going to be used.

Even if the review is not presented in a single chapter and you integrate your citations fairly evenly throughout the thesis, it is still helpful to make it clear to your reader how and where you address the various issues which have come from your reading. If integrating the literature in this continuous way, an overview of your approach in the introduction of the dissertation or thesis as well as appropriate headings and subheadings in the different chapters throughout the text is recommended.

For the more conventional and still more common practice of writing your literature review in one or more distinct chapters, it is strongly recommended that you include the following:

- an introduction which explains how your review is organised;
- headings and subheadings that provide a map to show the various strands of your argument;
- a summary where the key arguments are reiterated in a concise way.

If the review is long, as is likely for a PhD thesis, summaries interspersed throughout the chapter are helpful; these explain what you have argued so far and how this connects with what follows. An example of a 'transition statement' between sections is given in Example 6.1. The researcher refers to

what she has discussed and makes a link with the relevant literature that she is going to address next. The italics indicate the signalling language which informs the reader about how the text is organised.



Example 6.1 The transition between sections

eg → *Having defined* comfortable intelligibility as a target in pronunciation for learners, *the next question to be addressed is* how we can help learners to achieve this. The teaching profession believes that instruction does help; however an intuitive belief is not reason enough to plan a course of instruction. *The next section will look briefly at* research on the effects of second language instruction in general *and then turn to* the area of pronunciation instruction in particular.

Source: Moore, 2001: 5

Although it is impossible to prescribe a uniform structure for a literature review, there are some organisational principles which are followed by many research writers. You may draw on a combination of the different approaches mentioned below depending on which is most suitable for each particular part of your review.

Weissberg and Buker (1990: 45–6) propose three ways for ordering citations:

1 Distant to close
  Most distantly related to your work→Most closely related to your work
2 Chronological
  Earliest related work→Most recent related work
3 Comparison and contrast of different approaches or particular features or characteristics of relevant theories and research
  One approach→An alternative approach→Another approach

In relation to the first of these approaches, Rudestam and Newton (2007) talk about long shots, medium shots and close-ups to describe the different degrees of depth that you may go into when referring to source texts according to their closeness and relevance to your research. Long shots describe the references that provide the background context to the research. These references tend to be quite general, acknowledging that research has been done on the topic without going into detail. Medium shots are the references which have more bearing on the current research and although not critiqued in detail, enough information is given to show how they impact on the proposed